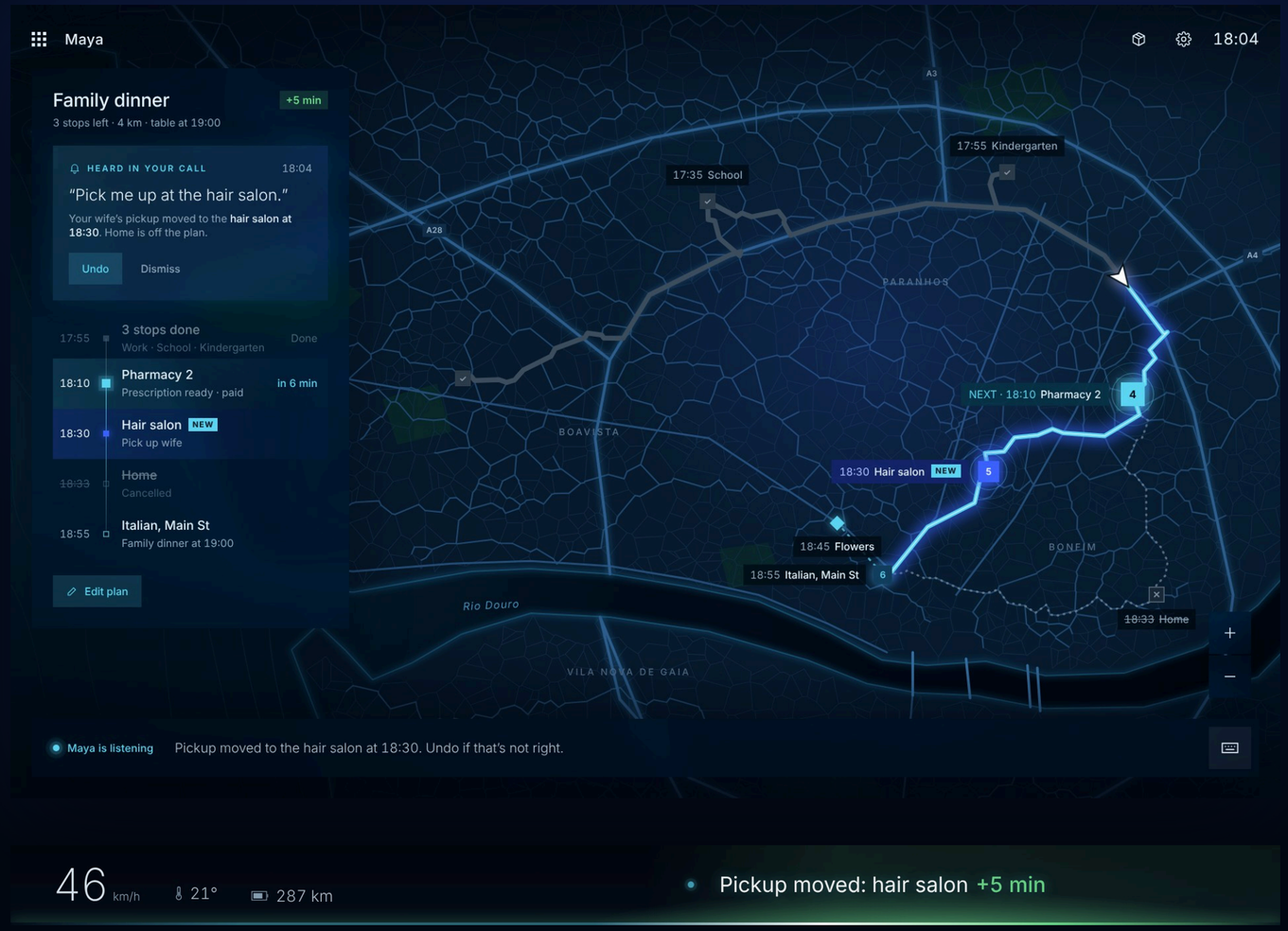
UX CHALLENGE · NEW AI ASSISTANT FOR INFOTAINMENT

Maya

An AI trip concierge for a Level 5 car

Michael shares his needs and goals. Maya plans the stops, phones the shops, pays and replans when the world, a conversation or he changes something. When nobody is in the car, she sends it to run his errands.



S17 · Plan change: ambient, with the LED strip at the same moment

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October 2026

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Variants and edge cases · validation and metrics · trade-offs and what's next

Every title and thumbnail here is a link. "Contents" at the bottom of each page brings you back.

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C5 p. 23
New errand



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Takeover request


V2 p. 28
Running late

WHAT I MADE, AND WHY

Each artifact answered one question before the next one started.

01	**Hypothesis and assumptions**	To commit to one bet and rank what could break it, so the riskiest parts (ambient intent, AI callers, an empty car) get designed and tested first.
02	**Hero script**	An assistant is mostly conversation. Writing every line first decided when Maya speaks, when she stays silent and what the screen must show instead.
03	**Journey map and task flow**	To see the whole evening at once: Michael's goals and feelings, what Maya does backstage, where the risks sit, and the logic behind each replan.
04	**Screen plan and strip specification**	To fix the screen inventory and IDs before drawing, and to decide what the passive LED strip may and may not show.
05	**Multimodal matrix**	To keep voice and touch equal: every voice action has a touch twin, every spoken line a caption.
06	**Edge cases and validation plan**	To design for the evening going wrong, and to say how each assumption would be tested and what success looks like.
07	**Decision log**	One source of truth while the concept changed, so names, IDs and rules stayed the same across every artifact.
08	**Greyscale wireframes**	To settle layout and hierarchy (plan on the left, map behind, Maya below) before any visual decisions.
09	**Mood board**	To choose a visual direction on purpose. Deep ocean won: in a dark cabin at night, light can carry meaning.
10	**Hi-fi screens and component inventory**	The whole story in final quality, built from one set of components so every state stays consistent.
11	**Clickable prototype**	To feel the timing of the story, and to run the remote tests in the validation plan.

THREE LINKS, ALL OPTIONAL

Plain web pages, no sign-in. This PDF stands on its own; the links add full-size screens, the working files and the prototype.

PROCESS

oliveiraanibal.github.io/maya/

Every working artifact listed here, in the order it was made.

SCREENS

oliveiraanibal.github.io/maya/screens/

All hi-fi screens at full size, the LED strip states and the component inventory.

PROTOTYPE

oliveiraanibal.github.io/maya/prototype/

The clickable story, S01 to C3, with the branches.

WORKING ALONE

A solo exercise with no user research. The assumptions are written down, ranked by risk, and paired with the tests that would check them.

THE PROBLEM AND THE BET

A self-driving car frees your hands, not your evening.

Navigation handles routes, not commitments. Even when the car drives, the passenger still works out the order of the stops, phones businesses and replans by hand whenever something changes, so the freed-up time is not really free.

INSIGHT

People think in needs, not addresses

“Family dinner tonight. Let’s get the kids and my wife, and I need that new prescription.” One sentence holds three pickups, a pharmacy to find and pay, a restaurant and a deadline.

VISION

“I share my needs and goals; everything else just happens, and I can see it happening.”

“With me in the car, Maya runs my trip. Without me, my car runs my errands.”

HYPOTHESIS

Time-poor people will hand trip logistics to an AI that **asks only what it cannot know**, acts on its own, follows the conversation and makes every action **visible and reversible**.

We are right when people arrive on time, rarely undo what Maya did, and trust her with more over time.

WHO

Michael, a father of two with a packed evening. Maya belongs to him, not to the car: she is in any car he signs in to, his own or a robotaxi.

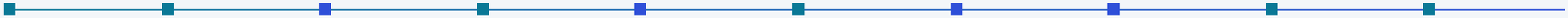
ASSUMED, NOT DESIGNED

Payments from Maya Wallet up to a limit he sets. Schools release the children to the car. The florist delivers to the restaurant.

ACCESSIBLE FROM THE START

Every task works with voice alone or touch alone. Everything Maya says is captioned, and colour never carries meaning on its own.

THE HERO SCENARIO: THE FAMILY DINNER RUN, ONE EVENING IN TEN SCENES



17:15	17:16	17:19	17:22	17:24	17:33	18:03	18:32	18:55	18:58
Sign-in	Briefing	Pharmacy calls	Flowers	The plan	Pickups	Wife’s call	The errand	Arrival	Courier run
		Reroute 2 · the world		Reroute 1 · the plan		Reroute 3 · the conversation	Courier errand		

THE BIGGER BET: COURIER MODE

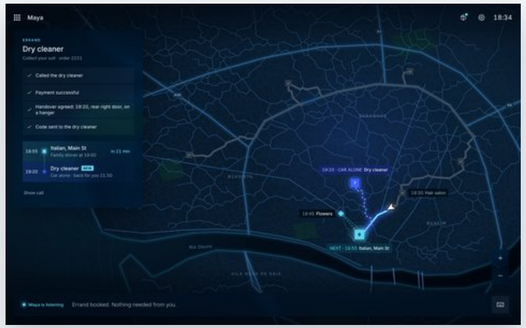
When the family is at dinner, the empty car runs the errand.

After the hair salon pickup, a message says Michael’s suit is ready at the dry cleaner. Maya offers to collect it once the family is at the restaurant, and to be back for them at 21:30. The car is the same one all evening, so the suit stays in it until Michael calls it.



S18 · Errand suggestion

A message says the suit is ready. Maya offers to collect it with the empty car.



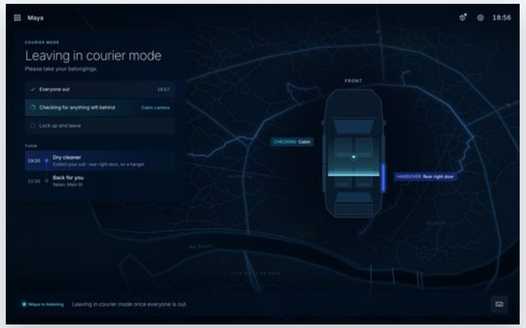
S19 · Errand booked

She calls, pays and agrees the handover. Only then is the errand booked.



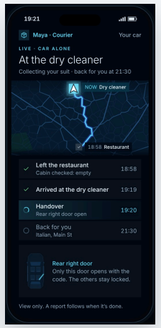
S20 · Arrival report

The arrival report ends with what happens next and when the car will be back.



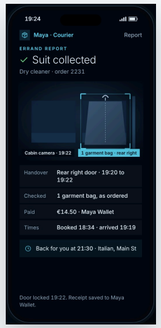
C1 · Leaving in courier mode

The car checks that the cabin is empty and nothing is left behind.



C2 · Phone: courier run

Michael follows the run on his phone, view only.



C3 · Phone: errand report

A report with a photo, the receipt and the times.

THE RULES

Empty cabin only
Before leaving, the car checks for people, pets and forgotten items.

No handoff, no commitment
An errand is booked only once the shop confirms the time, the person and the handover.

One door, one code
Only the agreed door opens, once, with a one-time code; the cabin camera checks what goes in.

The return time is the promise
Michael always knows when the car will be back; errands that can’t make it are declined.

Every run is reported
Live on the phone, then a photo, the receipt and the times.

Charging is not an errand
It never appears in the plan. The car charges out of sight.

RISKIEST ASSUMPTIONS

A13 shops will hand items to an empty car · A14 people will send it, and accept a cabin camera · A15 an empty Level 5 car may legally run errands.

PRINCIPLES AND THE LEVEL 5 STANCE

Nine rules every screen follows, in a car nobody has to drive.

1 Ask only what she can’t know

As many questions as needed, no more. Schools, pickups and the prescription come from the profile and messages.

S06

2 The plan follows the conversation

Ambient intent: what Maya hears in the cabin or a call can reshape the plan.

S17

3 Speak only when needed

The screen is the confirmation. Maya never reads the plan back.

S07 · S12

4 Do first, show clearly, undo easily

No approval gates and no countdowns. Every action lands in “Done for you”.

S09 · S14 · S17

5 The plan is alive

The world, the conversation or Michael can reshape it, and the ripple is visible.

S09 · S15 · S17

6 Spare time is the promise

Always visible. Green above 4 min, amber 0 to 4, red below 0, never colour alone.

S12 to S20 · V2

7 Tablet to act, strip to glance

The strip can’t be touched and carries nothing unique.

Strip

8 Voice and touch are equals

Every task works with either one alone.

S05 · S06 · S15

9 With you or without you

An empty car works for you, but only with an agreed handover.

S18 to C3

SAE DRIVING AUTOMATION LEVELS

0	1	2	3	4	5
No automation	Driver assistance	Partial	Conditional, takes over on request	High, in a defined area	Full, anywhere, no driver

WHY LEVEL 5

Only when nobody has to drive can an assistant use the whole trip, and can Michael talk, read and decide. One calm, trip-level experience instead of a design built around driver attention.

AT LEVEL 3 OR 4

The same concept could ship earlier, but every task would give way to a takeover request: Maya pauses and goes silent, and the tablet and strip count down together (V1, p. 28).

HOW MICHAEL STAYS AWARE

- **The map is always there.** Position, the route ahead and what’s done sit behind every screen.
- **The car states its intent first.** “Stop at School in 1 min”, where it pulls in and which doors unlock (S13).
- **Every change shows what and why.** Removed stops crossed out, new ones marked, the old route dotted (S09, S17).
- **Car data never leaves the strip.** Speed, temperature and range always hold its left half.
- **Nothing happens out of sight.** Calls are transcribed, payments listed, recent actions can be undone.

JOURNEY MAP · BEATS 1 TO 5

Emotion: Michael's expected feeling, 1 low to 5 high · A2 to A15: assumptions, p. 29

Setting off: from sign-in to a plan Michael can trust at a glance.

PHASE	Setting off				
TIME	17:15	17:16	17:19	17:22	17:24
BEAT	**Boarding and sign-in**	**Briefing**	**Pharmacy search**	**Flowers**	**The plan appears**
MICHAEL'S GOAL	Get going after work with no setup	Say what the evening needs, not where to go	Get the prescription as soon as possible	Not arrive empty-handed	Trust the plan at a glance
MICHAEL DOES OR SAYS	Gets in; signed in by face	"Family dinner tonight. Kids, my wife, and that prescription." Answers: the usual Italian on Main Street, at 19:00	Rides; may read the transcripts	"Yes. White roses."	Glances at it
MAYA SAYS	"Hi, I'm Maya, your concierge for the trip. How can I assist you?"	Asks only where, and what time	Silent	Suggests flowers once the pharmacy is secured, then silent	Silent
TABLET SHOWS	Sign in, then Welcome and "Going home" (S01, S02)	The "New plan" card fills in (S04 to S07)	Two calls with holds, then "Plan updated" (S08, S09)	Suggestion, then call, payment and "Delivery committed" (S10, S11)	Plan overview, spare time +5 min (S12)
STRIP SHOWS	◉ Home 17:48	◉ listening	Pharmacy changed · +5 min	◉ Calling Main St Florist, then "Flowers confirmed"	School 17:35 · +5 min
MAYA BACKSTAGE	Recognises Michael; predicts home from habit; loads calendar, contacts, messages	Parses needs; reads the prescription from messages; asks the gaps	Calls pharmacies on the route; pays; reroutes to pharmacy 2	Finds the birthday; picks an open florist near the restaurant; calls and pays from Maya Wallet	Builds the route; arrival 18:55; replaces the default route
EXTERNAL	none	none	Pharmacy 1: no stock. Pharmacy 2: ready in 10 min	Florist agrees delivery by 18:45	none
EMOTION					
RISK	Wrong destination guess (A2); face not recognised	Misheard or missing detail (A6)	No pharmacy has stock (A4)	Florist refuses an AI caller (A4, A7)	Too much detail (A8)
DESIGN RESPONSE	The guess is a quiet default; PIN or password as fallback	The card is the confirmation; tap Maya's bar to type	Silent reroute with a visible banner	Self-introduction and live transcript	Spare time first, details on tap

JOURNEY MAP · BEATS 6 TO 10

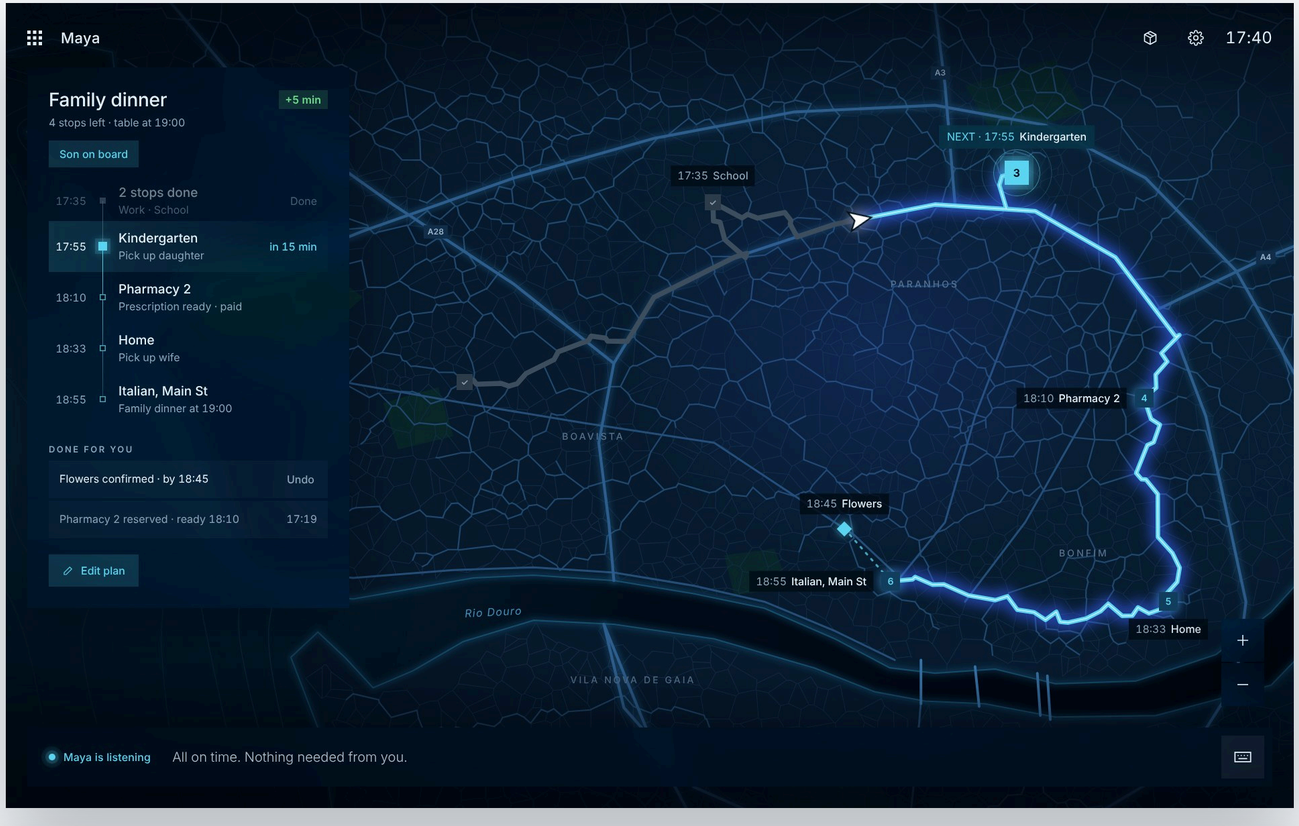
On the way, arrival, and the car alone.

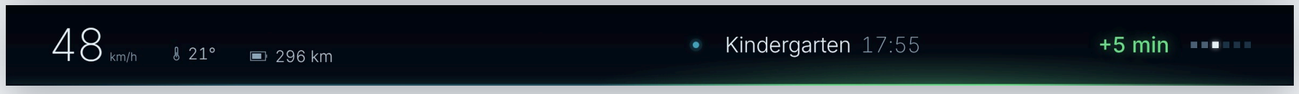
The last three rows are the service blueprint: backstage, the outside world, and how the design answers each risk.

PHASE	On the way			Arrival	Car alone
TIME	17:33	18:03	18:32	18:55	18:58
BEAT	**Pickups**	**Wife's call**	**The errand**	**Arrival**	**Courier run**
MICHAEL'S GOAL	Collect the kids without managing it	Absorb a change while on the phone	Get the suit without another trip	Arrive on time	Know the errand is done without watching it
MICHAEL DOES OR SAYS	Rides	Takes the call: "Sure, we'll be there."	"Yes, please."	Steps out with the family	At dinner; glances at the phone
MAYA SAYS	Silent	Silent, chime only	Offers to collect the suit with the empty car after the drop-off	"We're here, five minutes early. I'll collect your suit and be back for you at 21:30."	Silent; notifications only
TABLET SHOWS	What the car is about to do, pickup states, "Done for you" (S13, S14)	Heard phrase, crossed and new stops, Undo (S16, S17)	Errand suggestion, then call, payment, agreed handover and a new plan row (S18, S19)	Arrival report, then "Leaving in courier mode" (S20, C1)	Tablet off; phone shows the live run and the report (C2, C3)
STRIP SHOWS	Next stop updates	Pickup moved: hair salon · +5 min	◉ Calling the dry cleaner, then "Errand booked: dry cleaner"	Arrived 18:55 · +5 min, then "Courier · back 21:30"	Off (cabin empty)
MAYA BACKSTAGE	Tracks pickups and ETAs; tells the car where to stop and which doors to unlock	Hears the call; infers the change; replans	Reads the dry cleaner's message; checks the car is free after 18:55; calls, pays and agrees the handover; texts a one-time code	Confirms arrival; checks the cabin is empty and nothing is left behind	Drives to the dry cleaner; opens one door for the code; the camera checks the suit; reports; charges near the restaurant until Michael calls
EXTERNAL	School and kindergarten release the children	Wife asks to be picked up at the hair salon	Dry cleaner agrees to hang the suit in the rear right door at 19:20	Flowers delivered 18:44; prescription collected 18:10	Staff type the code and hang the suit at 19:20
EMOTION					
RISK	Child not at the pickup point	Ambient false positive (A3, A9)	Shops won't serve an empty car (A13); trust in sending it (A14)	Someone or something left in the cabin (A14)	Shop not ready or wrong item (A13, A14)
DESIGN RESPONSE	The intent card says how long the car waits	Chime, heard phrase, Undo with no countdown	No handoff, no commitment: booked only once the handover is agreed	Courier mode starts only with an empty cabin	One door, one code; a report with photo, receipt and times

SURFACES AND CHANNELS

A tablet to act, a band of light to glance, and the phone only when the car is alone.





TABLET · ABOUT 15″, 1920 × 1200

Everything interactive. 16:10, seen from about 70 cm. The plan on the left, the live map behind, Maya's bar at the bottom.

LED STRIP · BASE OF THE WINDSCREEN

Passive and always dark. Car data on the left, Maya on the right. A copy of the tablet, never the only channel. Off when the cabin is empty.

CHANNELS

V — Voice
Always listening in the cabin, no wake word. Only Michael's voice gives commands; other voices are context.

T — Touch
Everything on the tablet: tap, drag, type. Every tap area is at least 76 × 76 px.

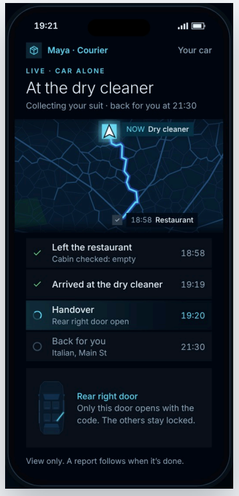
T — Keyboard
Tap Maya's bar to type. A large keyboard takes the lower half of the screen.

S — Captions
Everything Maya says appears in her bar, so nothing is audio only.

S — Chime
One soft sound, only for changes heard in conversation, always with the heard phrase on screen.

S — Light
The strip's horizon line carries the spare-time colour and flashes for 4 s on events.

V voice T touch S system output

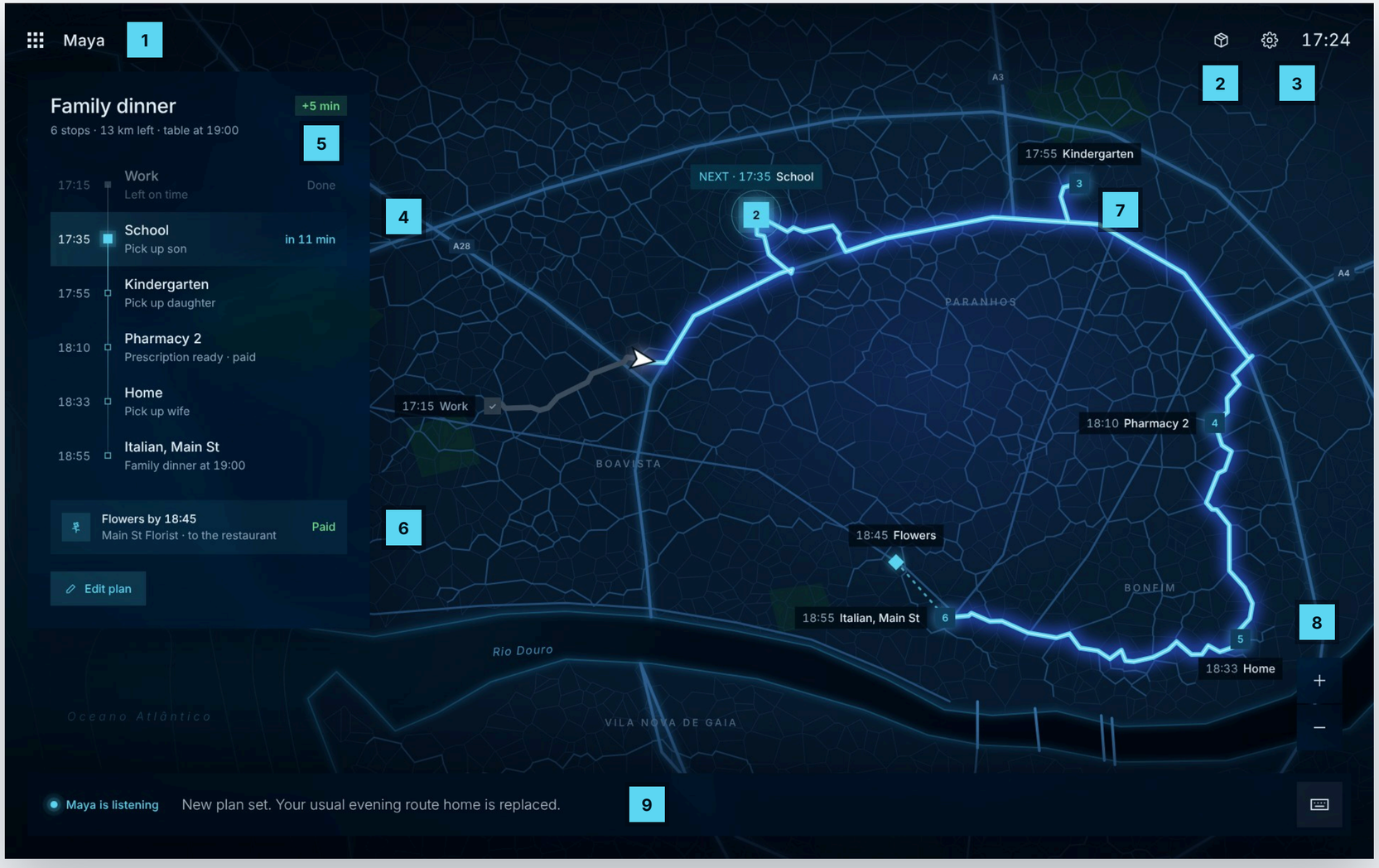


PHONE

Courier mode only, and view only: the live run (C2) and the report (C3).

SCREEN ANATOMY · S12 PLAN OVERVIEW

One layout everywhere: the plan on the left, the map behind, Maya below.



1 App launcher and Maya
Other apps sit behind the launcher. “Maya” names what is on screen.

2 Errands
Opens Errands (C4). A dot shows while an errand is booked or running.

3 Settings and clock
Nothing else in the top bar.

4 Plan panel
Trip title, stops left, distance and the table time; the stop timeline, next stop highlighted; Edit plan. Glass over the map.

5 Spare time
The promise, always visible: +5 min in green.

6 Done for you
What Maya arranged: flowers by 18:45, paid.

7 Live map
The whole trip. The route ahead glows, the travelled route is grey; square pins; sonar rings on the next stop.

8 Zoom
In and out, aligned with the keyboard button.

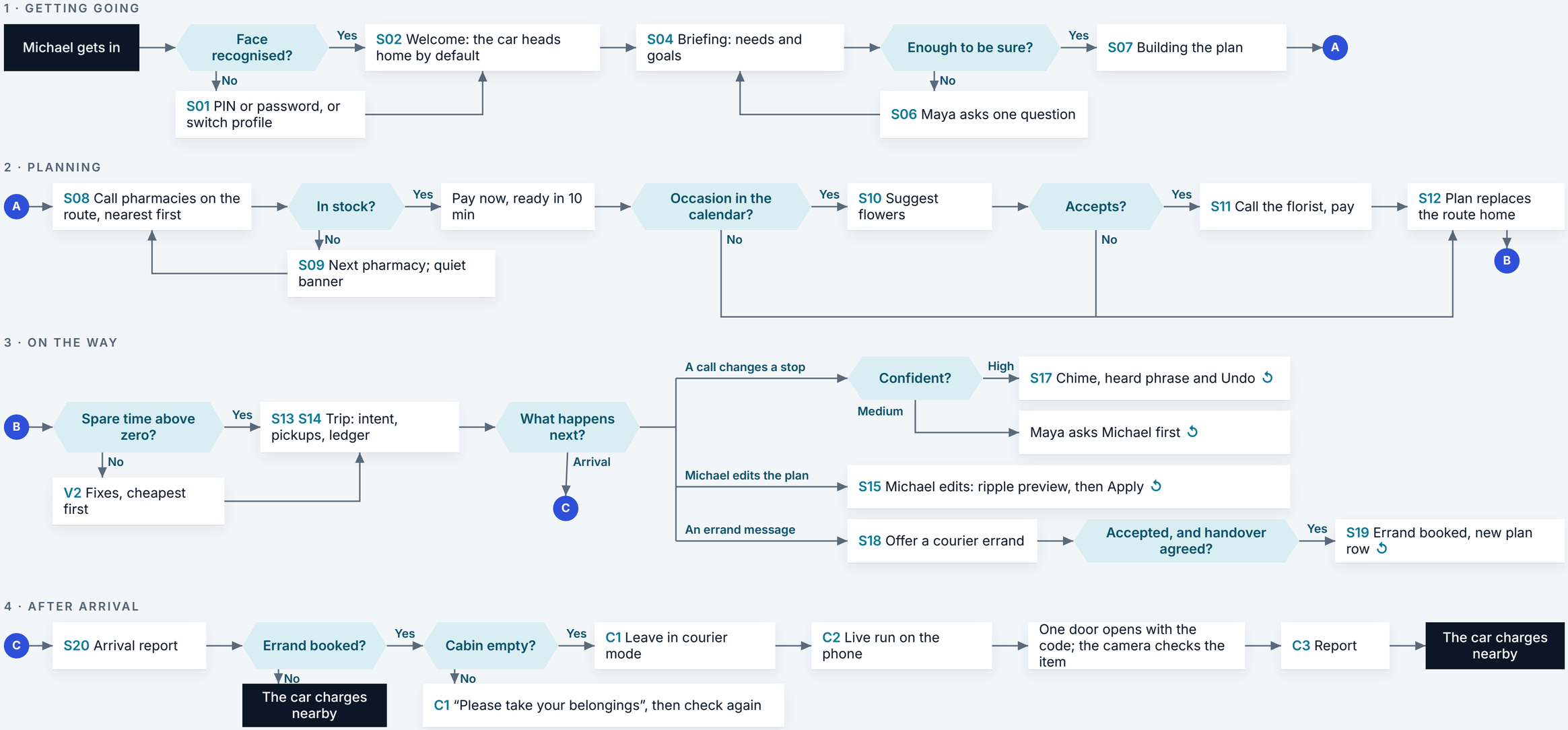
9 Maya’s bar
Her status (listening, thinking, speaking, on a call), her words as a caption, and the keyboard.

The map is Porto, drawn from its real coast, river and main roads; the minor streets are illustrative.

TASK FLOW

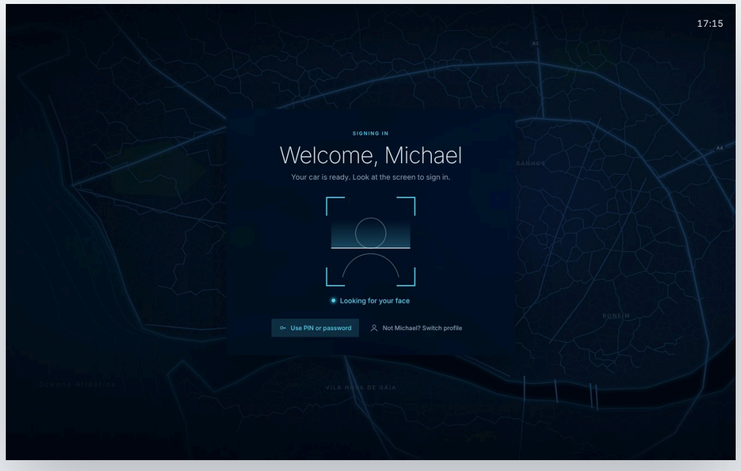
Every decision behind the evening, in four lanes.

step · decision · continues · back to the trip



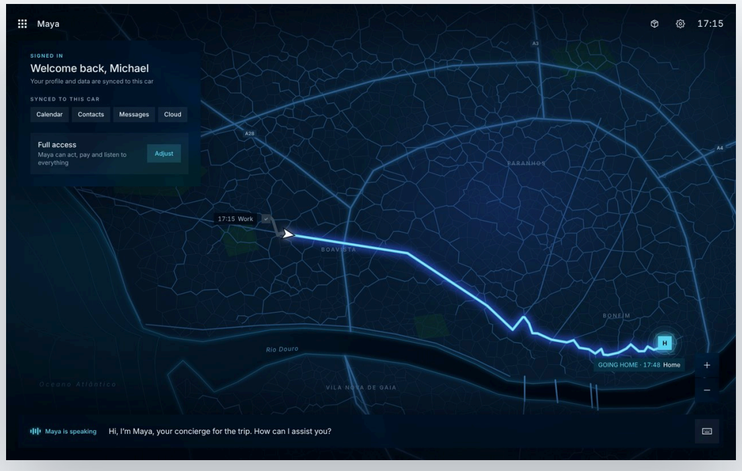
SCENE 1 · 17:15 · BOARDING AND SIGN-IN

The car expects Michael, signs him in by face and sets off towards home.



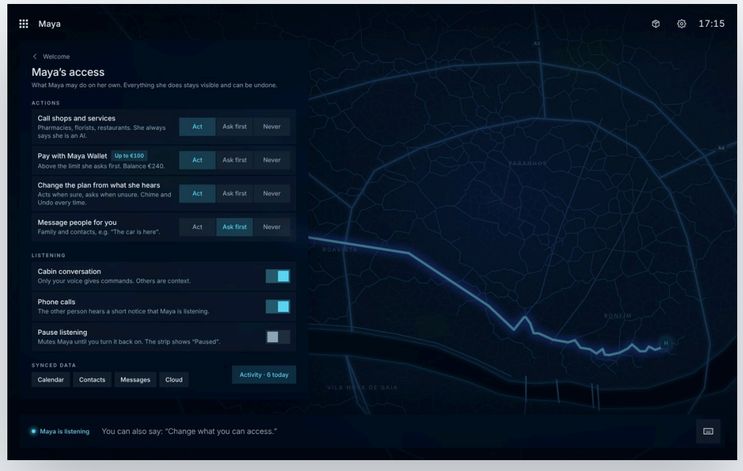
S01 Sign in

- The car expects Michael but signs him in only once it recognises his face.
- Fallbacks: Use PIN or password, or “Not Michael? Switch profile”.
- Before sign-in the top bar shows only the clock.



S02 Welcome

- “Welcome back, Michael”: calendar, contacts, messages and cloud synced to this car.
- Already moving on the most probable route: Going home, 17:48.
- “Full access” sums up Maya’s rights, with Adjust.



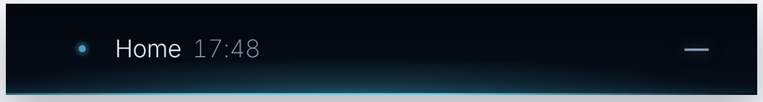
S03 Maya’s access

- Opened from Adjust. Per action: Act, Ask first or Never.
- Maya Wallet pays on its own up to €100; above that, Maya asks first.
- Listening: cabin, phone calls (the other person hears a notice), pause.

SPOKEN

MAYA	“Hi, I’m Maya, your concierge for the trip. How can I assist you?”
TOUCH TWIN	Adjust opens Maya’s access; by voice, “Change what you can access.”

STRIP · MAYA’S HALF



Home 17:48. No plan yet, so spare time is not shown and the line stays cyan.

SCENE 2 · 17:16 · BRIEFING

Michael says what the evening needs. Maya asks only what she can’t know.

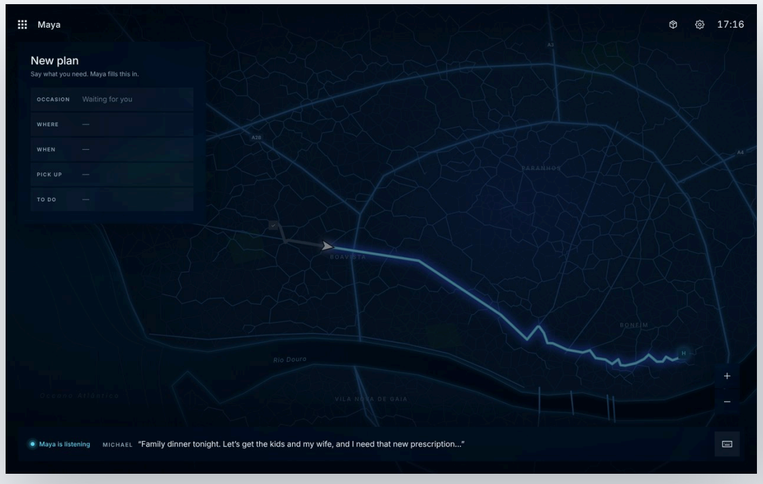
SPOKEN

MICHAEL	“Hi Maya. Family dinner tonight. Let’s get the kids and my wife, and I need that new prescription from my messages as soon as possible.”
MAYA	“Where is the dinner?”
MICHAEL	“The usual Italian on Main Street.”
MAYA	“What time?”
MICHAEL	“Seven.”
TOUCH TWIN	Type in Maya’s bar (S05) and answer in the search field (S06).

STRIP · MAYA’S HALF

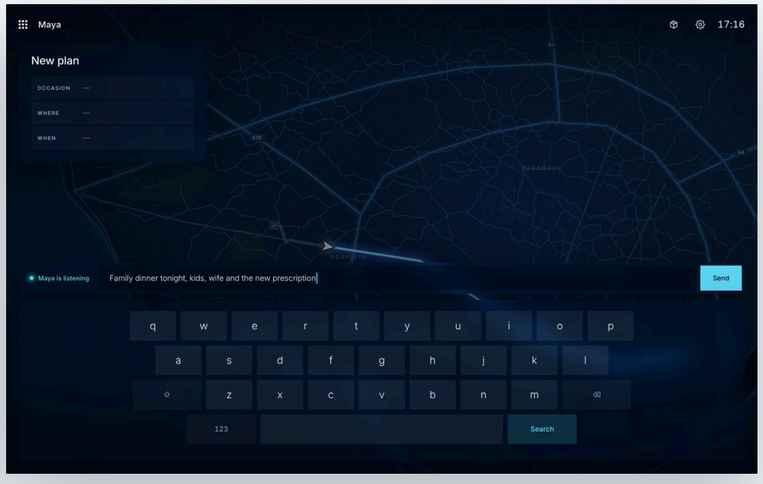


The presence dot brightens while Maya hears Michael. Home 17:48.



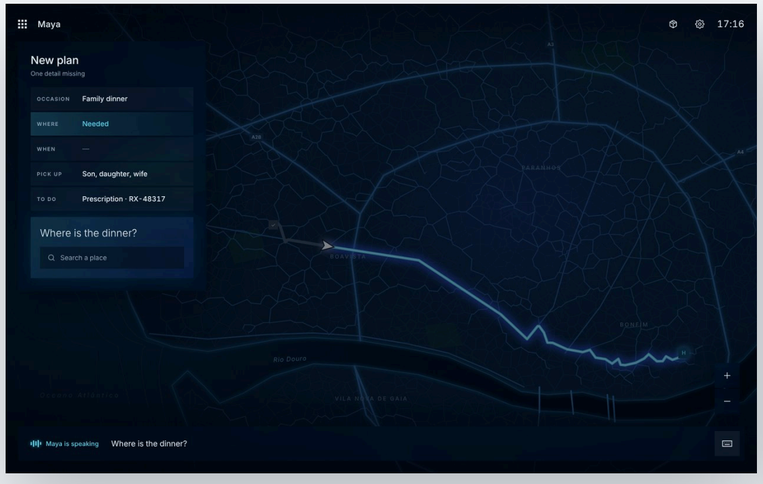
S04 Briefing: listening

The “New plan” card: occasion, where, when, pick up, to do. It fills in as Michael talks.



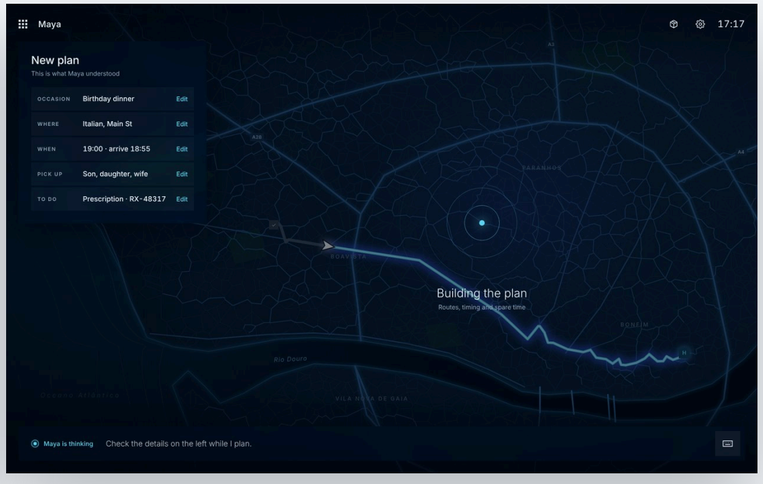
S05 Briefing: keyboard

Touch twin: tap Maya’s bar to type. The keyboard takes the lower half; the card shrinks.



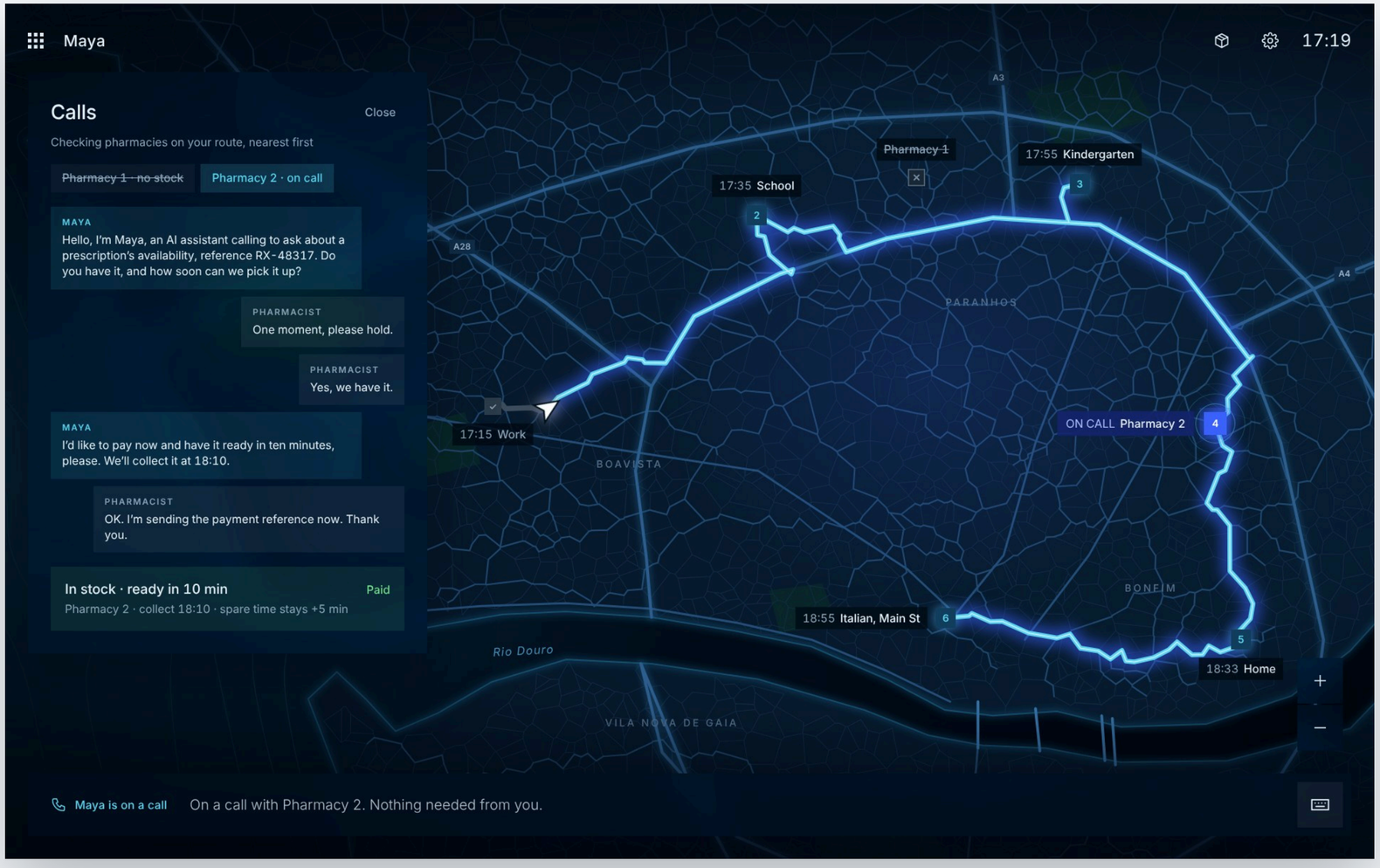
S06 Briefing: Maya asks

Maya knows the pickups and the prescription. She asks only where, then when. No suggested answers.

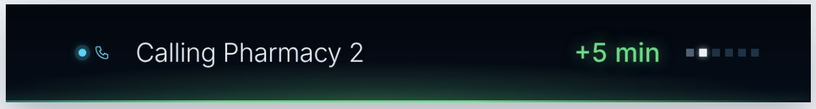


S07 Briefing: building the plan

“This is what Maya understood”, all still editable, while she builds the plan. No read-back.



STRIP · MAYA'S HALF



On a call: dot and phone. The plan now exists, so spare time appears: +5 min.

SPOKEN

MAYA	"Hello, I'm Maya, an AI assistant calling to ask about a prescription's availability, reference RX-48317."
PHARMACIST	"Yes, we have it."
MAYA	"I'd like to pay now and have it ready in ten minutes, please."

SCENE 3 · 17:19 · PHARMACY SEARCH

Maya phones pharmacies on the route, nearest first. The cabin stays quiet.

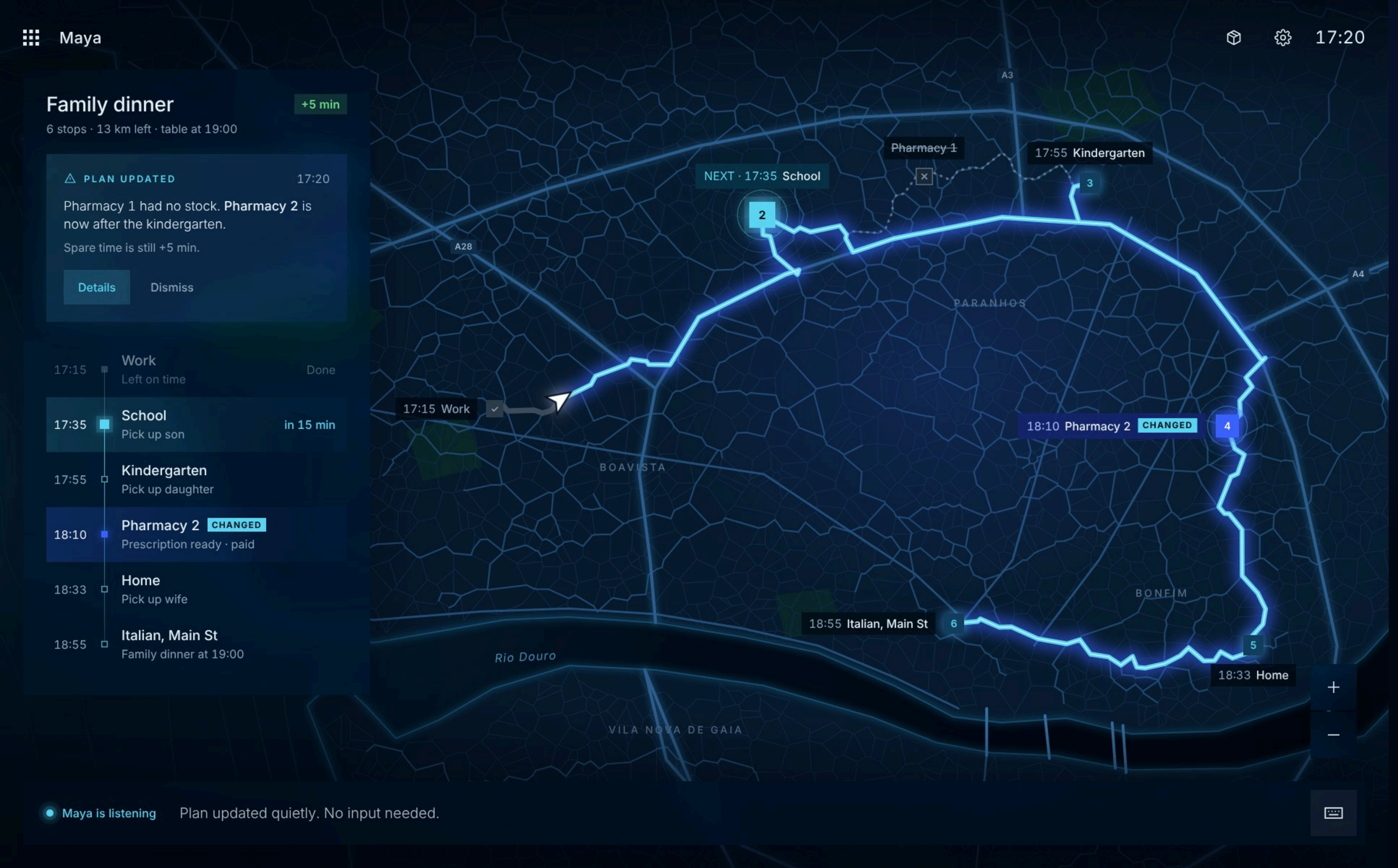
S08 Retailer calls

- Calls are silent in the cabin and transcribed live. Maya always says she is an AI.
- Pharmacy 1 has no stock and is struck through. Pharmacy 2 puts her on hold, then confirms.
- Paid from Maya Wallet, ready in 10 min, collected at 18:10. Spare time stays +5 min.
- On the map, Pharmacy 1 is crossed out and Pharmacy 2 marked "on call".
- The shared screen shows the prescription's reference number, never the medication.

TOUCH TWIN — Close hides the calls panel. Nothing is needed from Michael.

NEXT

The plan updates itself, quietly: S09 on the next page, as large as the page allows.



SCENE 3 · 17:20

S09

Plan change: automatic

A change in the world, made silently. Pharmacy 1 had no stock, so Pharmacy 2 now comes after the kindergarten.

No chime, no input. The banner says what changed and why, and stays until dismissed.

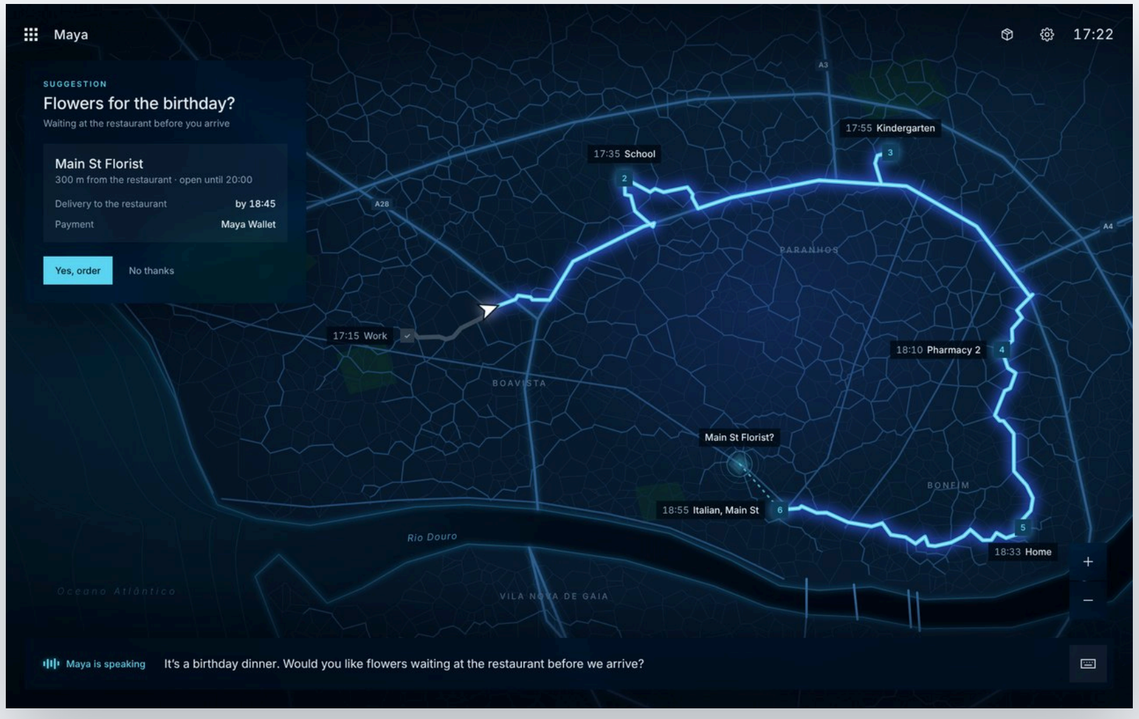
Pharmacy 2 is marked CHANGED in the list and on the map; the old route stays dotted. Spare time: still +5 min.

Strip: “Pharmacy changed +5 min” for 4 seconds.

CONTENTS
15

SCENE 4 · 17:22 · FLOWERS

A birthday in the calendar becomes one suggestion, then one call.

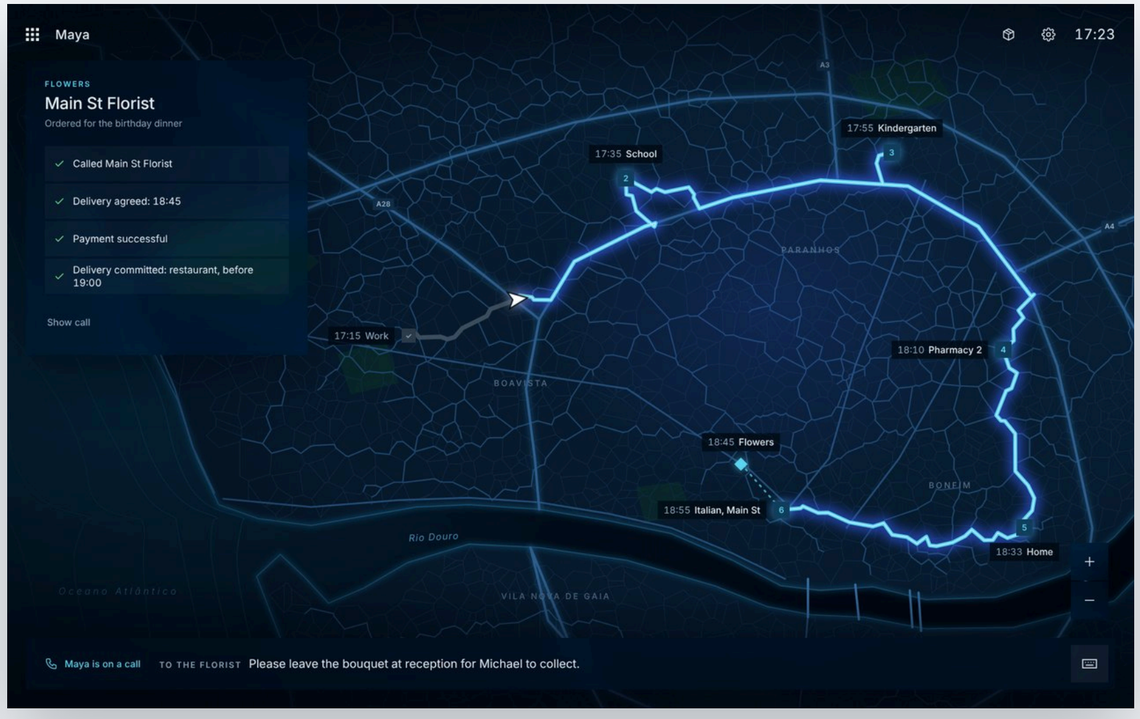


S10 Gift suggestion

- Offered once the pharmacy is secured. One florist, no bouquet choices until Michael says yes.
- Main St Florist, 300 m from the restaurant: delivery by 18:45, paid with Maya Wallet.
- Until he answers, the florist is a dotted pin on the map.

SPOKEN

MAYA	“It’s a birthday dinner. Would you like flowers waiting at the restaurant before we arrive?”
MICHAEL	“Yes. White roses.”
TOUCH TWIN	Yes, order or No thanks.



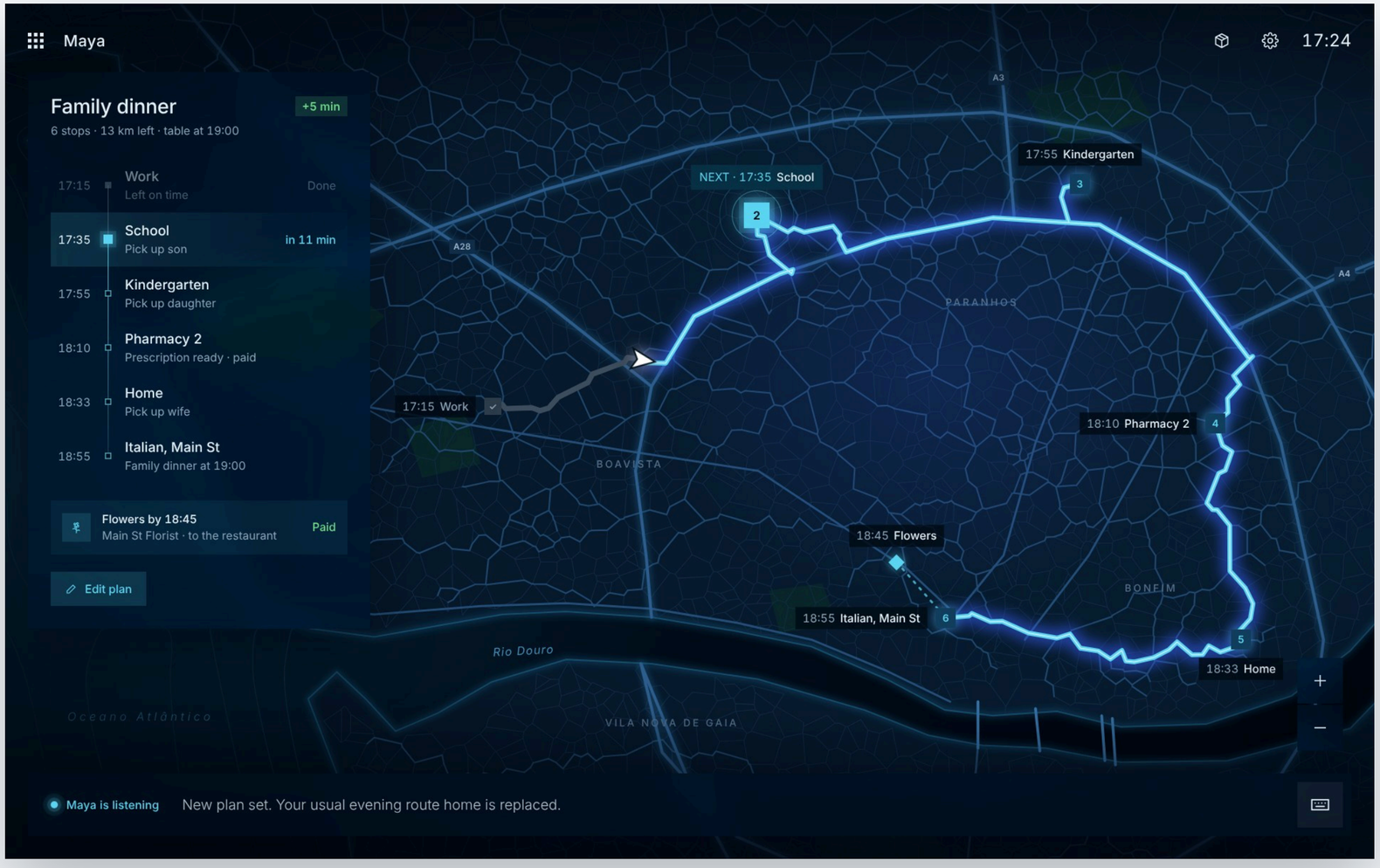
S11 Florist order

- Four confirmed steps: called, delivery agreed for 18:45, payment successful, delivery committed before 19:00.
- Show call opens the transcript; Maya’s last words to the florist stay in her bar.
- The pin turns solid, with a dotted delivery line to the restaurant.

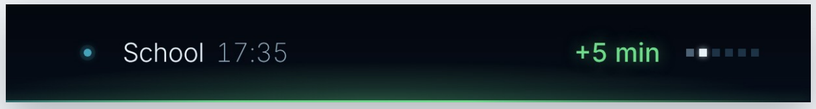
STRIP · MAYA’S HALF



Speaking: rings around the dot. School 17:35, +5 min.



STRIP · MAYA'S HALF



The resting state: next stop, spare time and one square per stop.

SCENE 5 · 17:24 · THE PLAN APPEARS

The route home is replaced by the plan. No speech: the screen is the confirmation.

S12 Plan overview

- Family dinner: 6 stops, 13 km left, table at 19:00, and +5 min of spare time at the top.
- The stop timeline: Work done and greyed, School next with "in 11 min".
- Flowers by 18:45, paid, in their own row; Edit plan below.
- The whole trip on the map: the route ahead glows, the travelled route is grey, the next stop has sonar rings.
- The car never stopped moving: the plan simply takes over from "Going home".

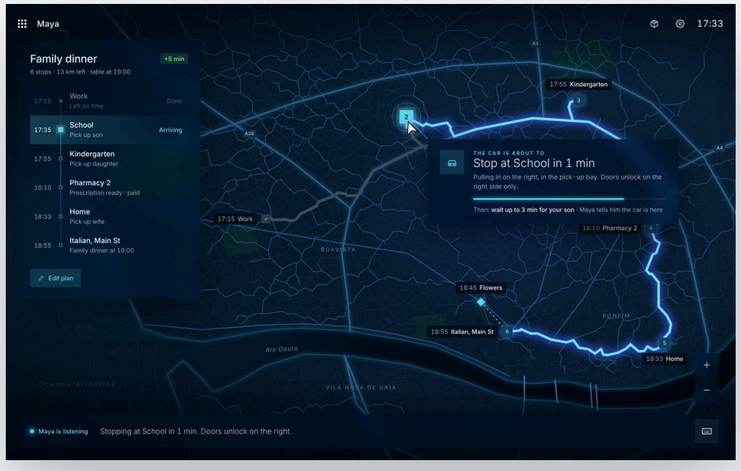
TOUCH TWIN — Tap a stop, Edit plan (S15), zoom. By voice: "What's next?"

SPOKEN

MAYA'S BAR — "New plan set. Your usual evening route home is replaced."

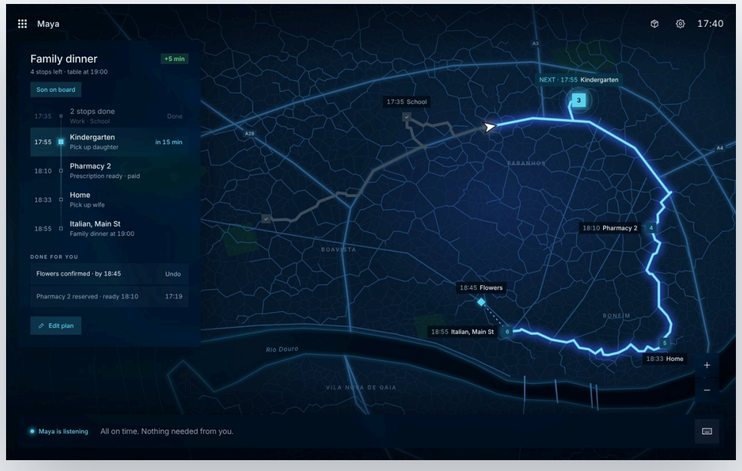
SCENE 6 · 17:33 · PICKUPS

The car says what it is about to do. Maya keeps a ledger of what she did.



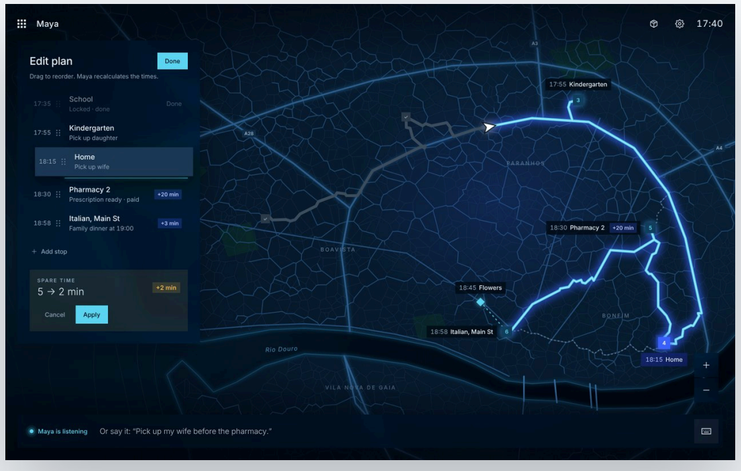
S13 Vehicle intent

- About a minute before each stop: where the car pulls in and which doors unlock.
- Then: wait up to 3 min; Maya tells his son the car is here.
- The stop row reads “Arriving”.



S14 Active trip

- “Son on board”; finished stops fold into one row.
- “Done for you”: the last two actions, with Undo on the latest only.
- Edit plan is always one tap away.



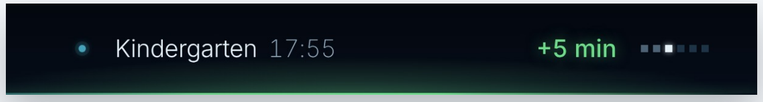
S15 Edit plan

- A branch: drag to reorder. The ripple shows every later time, +20 min, +3 min.
- Spare time 5 → 2 min, in amber. Nothing changes until Apply.
- The map previews the new route; the old one is dotted.

SPOKEN

MAYA’S BAR	“Stopping at School in 1 min. Doors unlock on the right.”
MAYA’S BAR	“All on time. Nothing needed from you.”
TOUCH TWIN	Edit plan, or say “Pick up my wife before the pharmacy.”

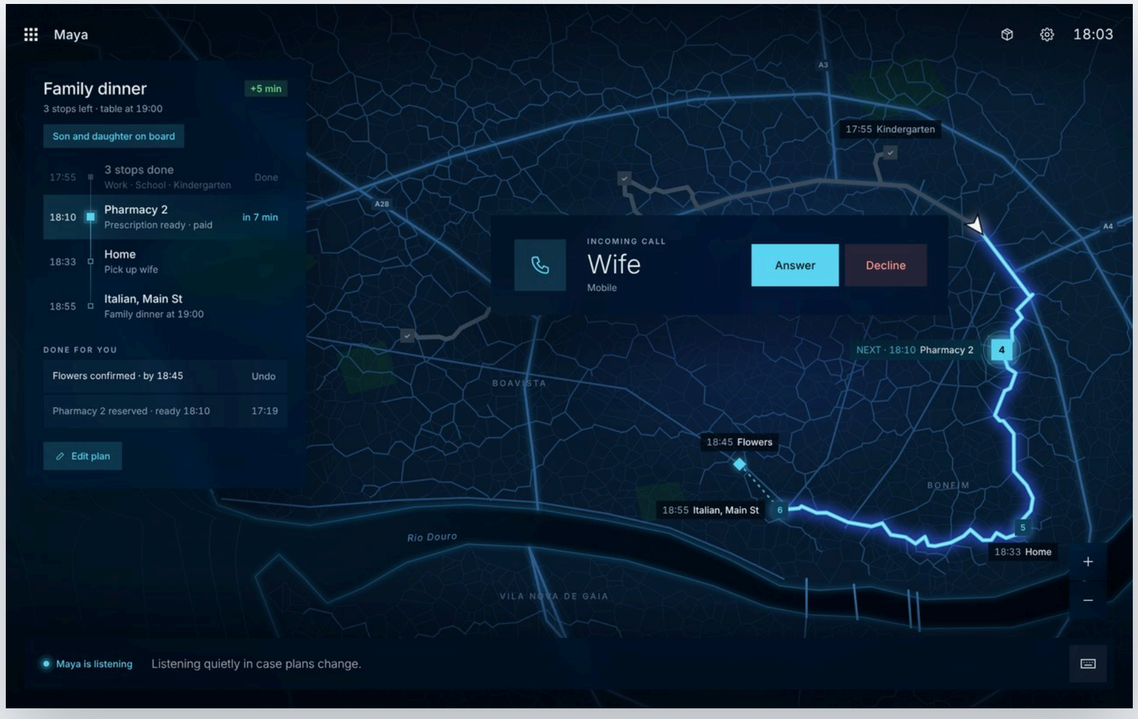
STRIP · MAYA’S HALF



The next stop updates as stops complete: Kindergarten 17:55, +5 min.

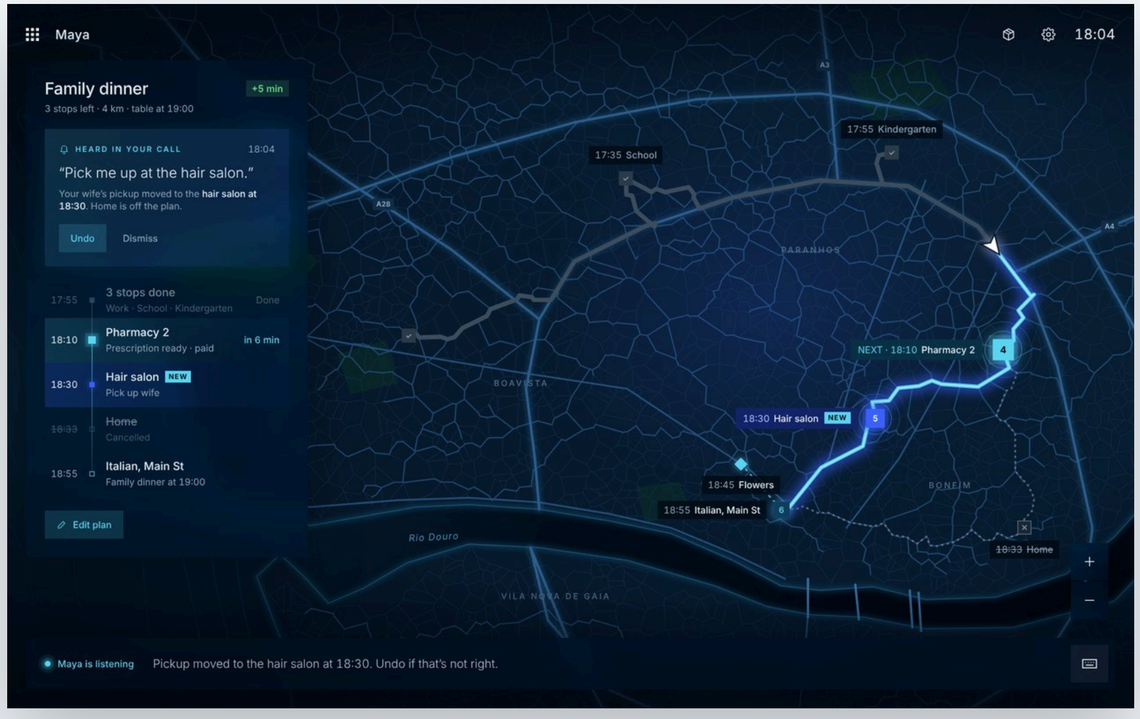
SCENE 7 · 18:03 · THE WIFE'S CALL

Maya hears “pick me up at the hair salon” and moves the stop, with a chime and Undo.



S16 Incoming call

- A call card over the map: Wife, Mobile, Answer and Decline. The plan stays visible behind it.
- Maya listens quietly; the other person hears a short notice that she is listening.



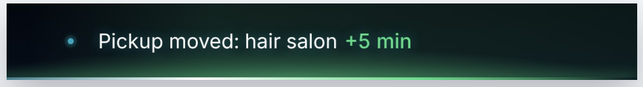
S17 Plan change: ambient

- “Heard in your call”: the phrase, what changed, then Undo and Dismiss. No countdown.
- Hair salon marked NEW at 18:30; Home crossed out; the old route dotted.
- One soft chime, no speech. If Maya were unsure, she would ask instead.

SPOKEN

WIFE	“Change of plan: pick me up at the hair salon, not at home.”
MICHAEL	“Sure, we'll be there.”
TOUCH TWIN	Undo, or change the pickup in Edit plan (S15).

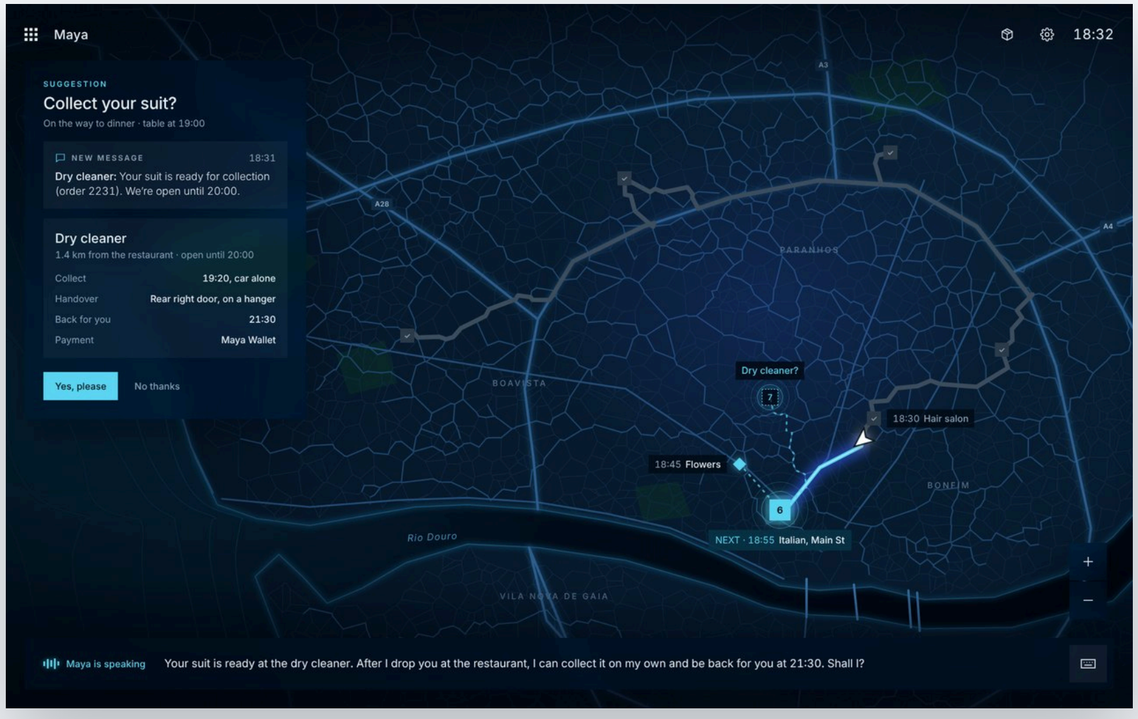
STRIP · MAYA'S HALF

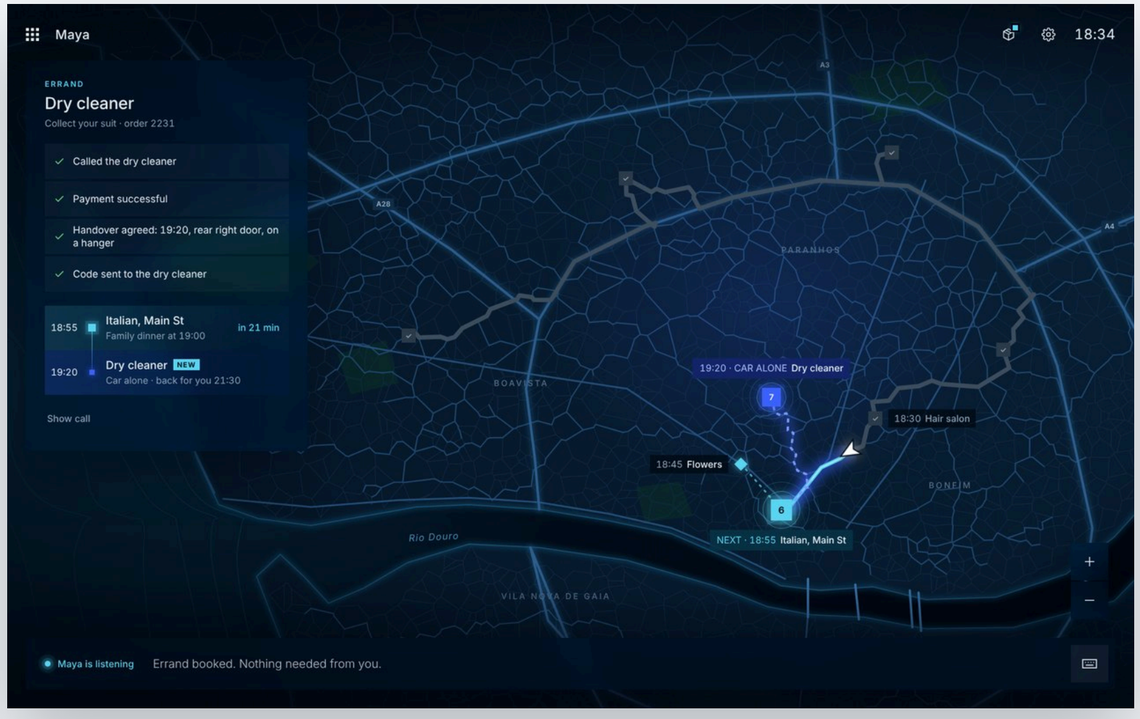


A 4-second event: “Pickup moved: hair salon +5 min”. The line flashes white.

SCENE 8 · 18:32 · THE ERRAND

A message from the dry cleaner becomes an errand for the empty car.





S18 Errand suggestion

- The dry cleaner's message is shown, not read aloud.
- The offer: 19:20, car alone, rear right door, back at 21:30, Maya Wallet.
- The dry cleaner is a suggested pin until Michael answers.

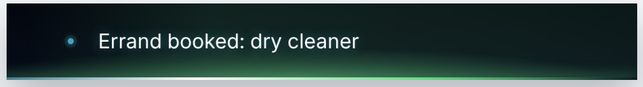
S19 Errand booked

- Booked only once the handover is agreed: called, paid, handover agreed, code sent.
- A new plan row after the restaurant: 19:20 Dry cleaner, car alone, back for you 21:30.
- A dot on the Errands icon while an errand is booked or running.

SPOKEN

MAYA	"Your suit is ready at the dry cleaner. After I drop you at the restaurant, I can collect it on my own and be back for you at 21:30. Shall I?"
MICHAEL	"Yes, please."
TOUCH TWIN	Yes, please or No thanks.

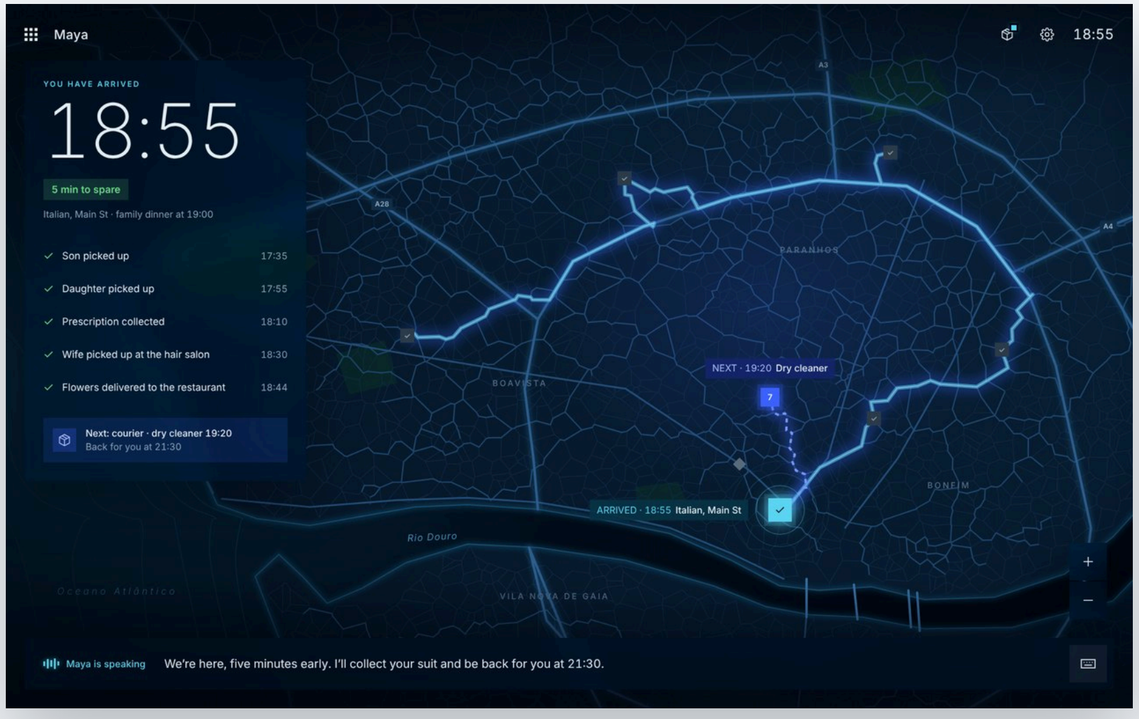
STRIP · MAYA'S HALF

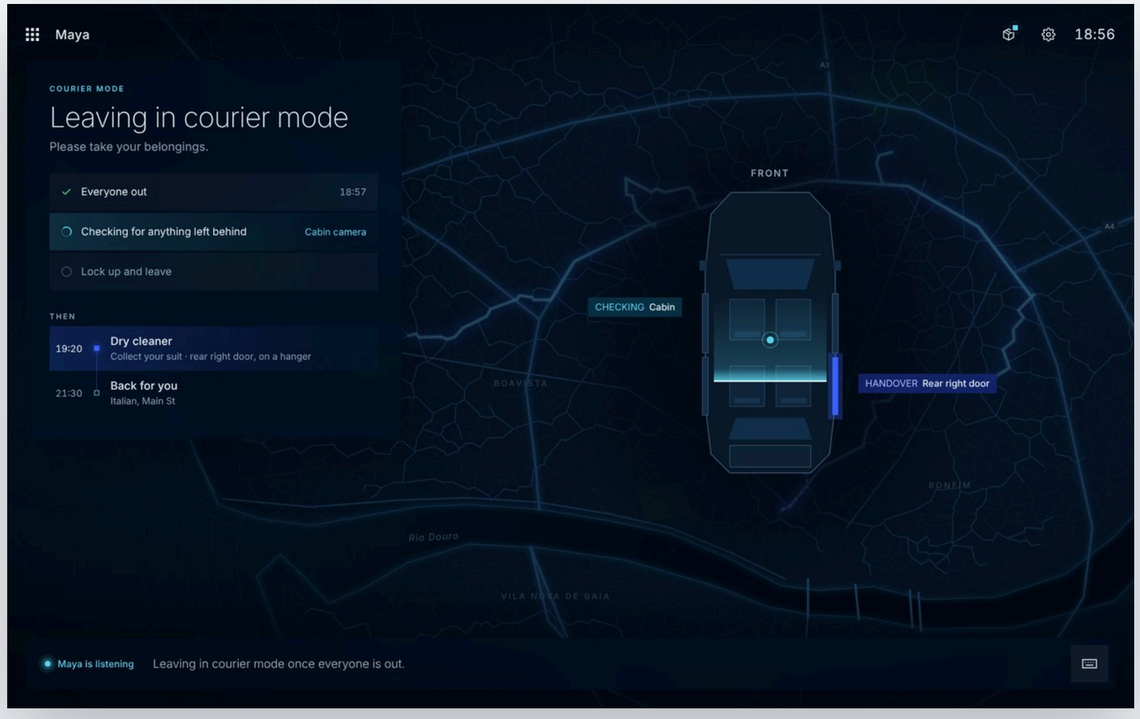


"Errand booked: dry cleaner" for 4 seconds. Spare time is unchanged.

SCENE 9 · 18:55 · ARRIVAL

Five minutes early. Then the car checks it is empty and leaves on its errand.





S20 Arrival report

- 18:55 and 5 min to spare, readable in one glance.
- Everything done, with times: son, daughter, prescription, wife at the hair salon, flowers delivered 18:44.
- Next: courier, dry cleaner 19:20, back for you at 21:30.

C1 Leaving in courier mode

- "Please take your belongings."
- Everyone out, then the cabin camera checks for anything left behind, then lock up and leave.
- A top view of the car shows the cabin check and the handover door.

SPOKEN

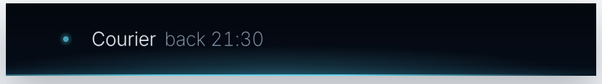
MAYA "We're here, five minutes early. I'll collect your suit and be back for you at 21:30."

STRIP · MAYA'S HALF

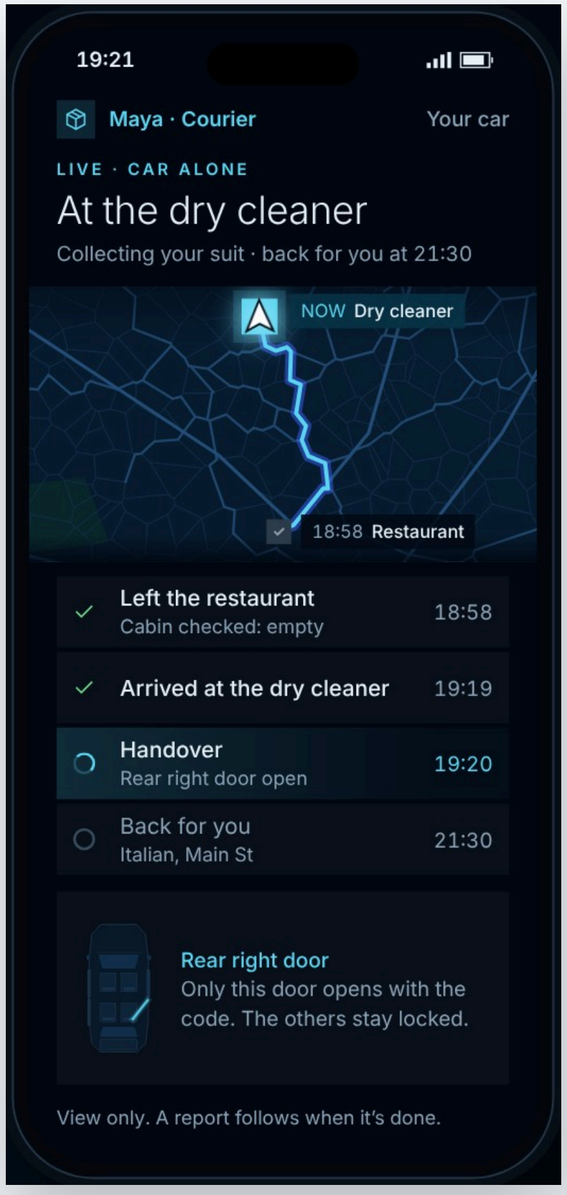


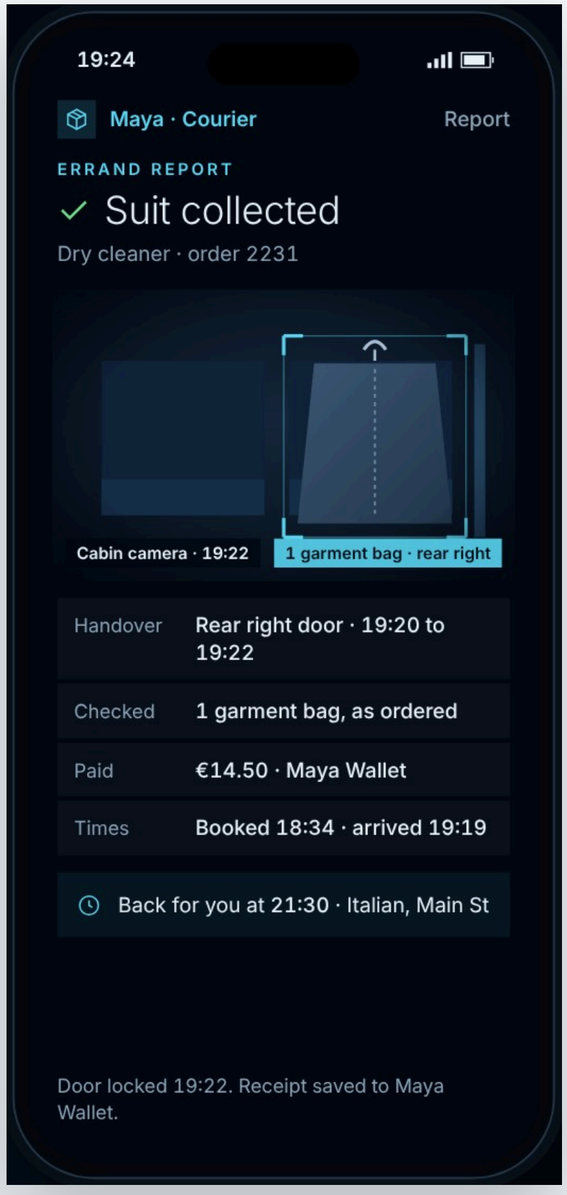
Arrived 18:55, every square filled, +5 min.

THEN



"Courier · back 21:30" while the family gets out.





SCENE 10 · FROM 18:58 · THE COURIER RUN

Michael follows the run on his phone, view only, and gets a report.

C2 Phone: courier run

- Live, with the car alone: the map and the steps. Left the restaurant 18:58, arrived 19:19, handover 19:20, back for you 21:30.
- Which door is in use: only the rear right door opens with the code; the others stay locked.
- View only: no controls and no voice.

C3 Phone: errand report

- A still from the cabin camera: 1 garment bag, rear right.
- Handover 19:20 to 19:22, checked as ordered, paid €14.50, booked 18:34, arrived 19:19.
- Back for you at 21:30. Door locked 19:22, receipt saved to Maya Wallet.

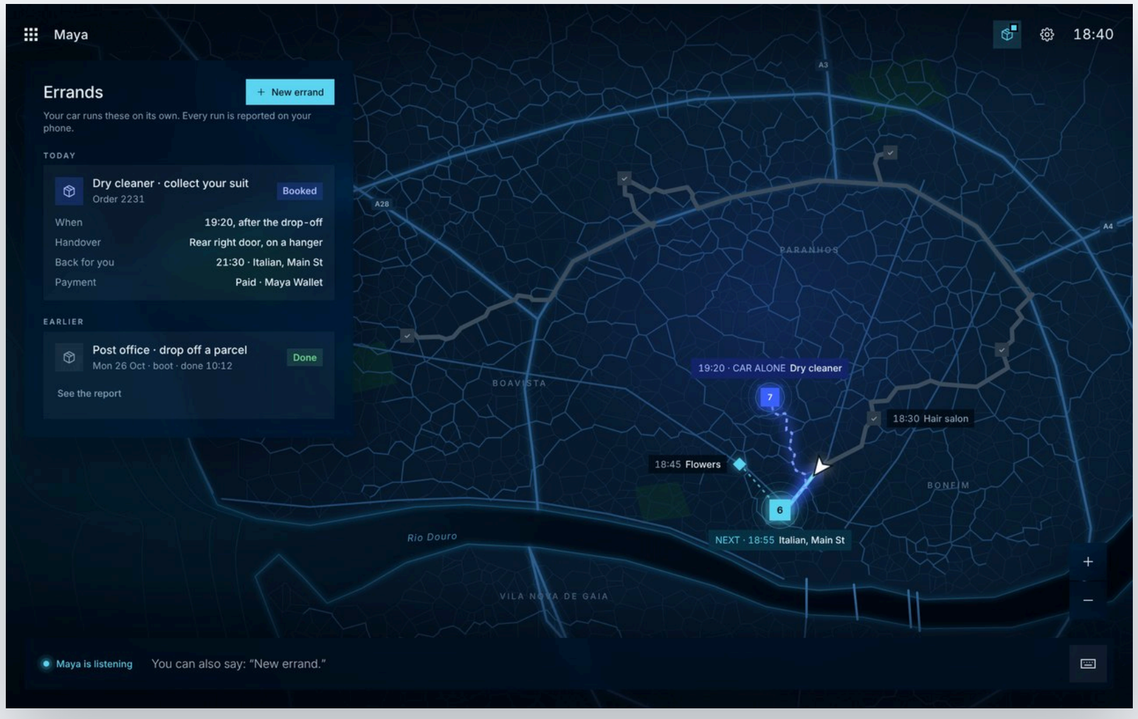
WHERE THE STORY ENDS

The car drives to a charging station near the restaurant. Charging is not an errand, so it never appears in the plan. The return at 21:30 is promised, not shown: the same car, with the suit, waits until Michael calls it, as he did at 17:15.

STRIP — Off: the cabin is empty, so the tablet and the strip are off.

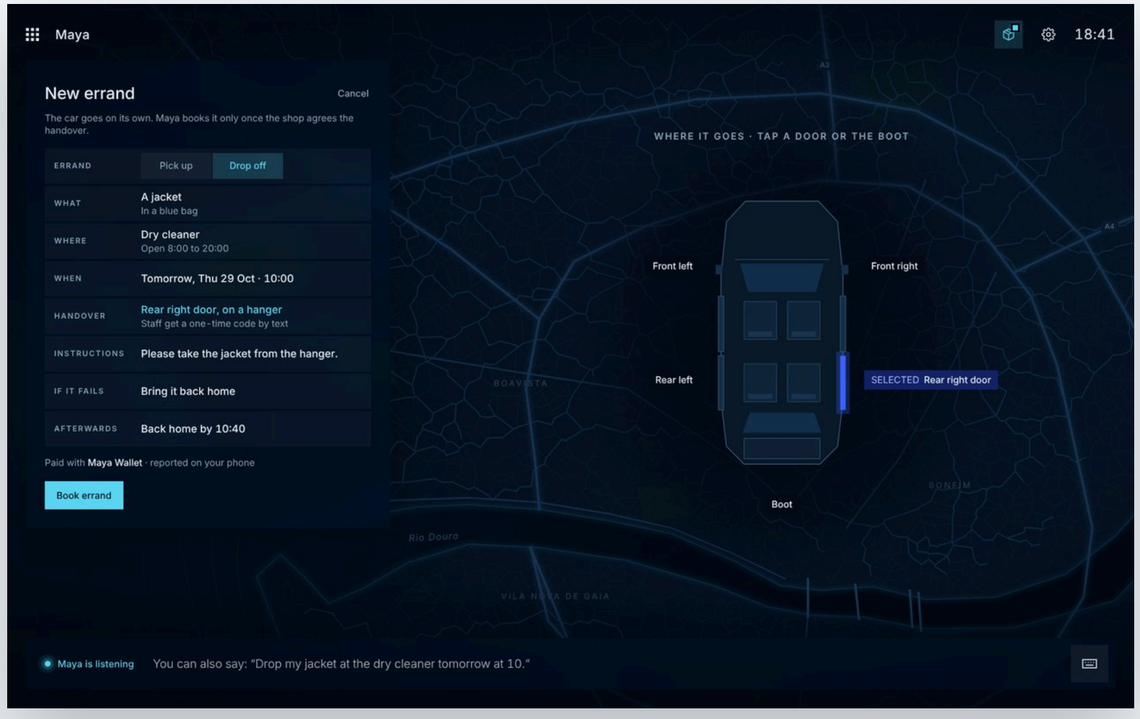
ANY TIME · ERRANDS BY HAND

Errands can also be booked by hand, from the box icon beside Settings.



C4 Errands

- Today: the dry cleaner, Booked, with when, handover, back for you and payment.
- Earlier: Post office, drop off a parcel, Mon 26 Oct, boot, Done, with its report.
- New errand at the top.



C5 New errand

- Pick up or Drop off; what, where, when.
- Handover: tap a door or the boot on a top view of the car. Staff get a one-time code by text.
- Instructions, what to do if it fails, when the car is back; then Book errand.

SPOKEN

VOICE TWIN "Drop my jacket at the dry cleaner tomorrow at 10."

DESIGN SYSTEM · DEEP OCEAN FOUNDATIONS

Our own tokens, with sizes checked against automotive guidance.



Abyss
Background

Glass
Panels, 30–60% over the map

Lumen
Primary accent

Cobalt
Glow, new items, errands

Green
More than 4 min spare

Amber
0 to 4 min spare

Red
Late, takeover

Text
Primary

Muted
Secondary

TYPE · INTER

Display	128 / ExtraLight	Arrival time (S20)
Hero	44–60 / Light	Sign-in, call card, courier title
Large	24–30 / Light	Clock, heard phrase, intent
Title	28 / Regular	Panel titles
Body	19–20 / Regular	Stops, Maya's captions
Secondary	16–17 / Regular	Times, details, notes
Label	13 / SemiBold caps	Kickers and row labels
Badge	12 / SemiBold caps	NEW, CHANGED, CHEAPEST

Dark first: the night cabin is the hard case, and light can carry meaning. The strip is always dark. Material 3 and Android for Cars are references for sizes and behaviour only.

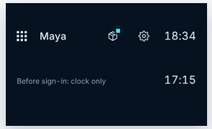
RULES

- **Sharp corners, no borders.** Fill, glass and spacing group things, never lines.
- **Glass over a full-bleed map.** Panels at 30 to 60% with a 24 px blur, so the map is always there.
- **76 × 76 px tap areas.** Visual sizes stay (48 to 64 px), but every tap area meets the Android for Cars minimum of 76 dp, at about 1 px per dp.
- **16 px minimum text.** Except uppercase labels (13 to 14 px) and small badges (12 px).
- **About 15″, 1920 × 1200, at about 70 cm.** A 16:10 tablet, like today's large infotainment screens.
- **Colour is never alone.** Spare time always has a number; late adds a minus sign and a warning mark.
- **One sound.** A soft chime, only for changes heard in conversation.
- **Motion with a job.** Strip events last 4 seconds; Maya's presence pulses slowly; nothing loops while she speaks.

COMPONENT INVENTORY · 1 OF 2

38 components. Chrome, map and content.

Specimens drawn with the same code as the screens.
Every tap area is at least 76 × 76 px.



CHROME

Top bar

App launcher · Maya wordmark · Errands button with badge · Settings · clock. Clock only before sign-in

All tablet screens

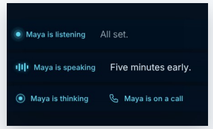


CHROME

Icon button

Default · active · with badge. 48 to 64 px visual, 76 px tap area

All tablet screens



CHROME

Maya's bar

Listening · thinking · speaking · on a call. Caption in one or two lines, with speaker label. Keyboard button; text-entry mode

S02 to S20, C1, C4, C5, V1, V2

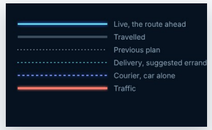


CHROME

Zoom control

Zoom in · zoom out

Map screens

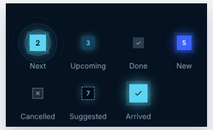


MAP

Route line

Live · travelled · previous plan · delivery · courier · suggested errand · traffic

S02 to S20, C1, C4, V2

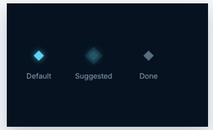


MAP

Stop pin

Upcoming · next with sonar rings · done · new · cancelled · suggested · arrived

S02 to S20, C4, V1, V2

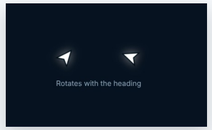


MAP

Delivery pin

Default · suggested · done

S10 to S20

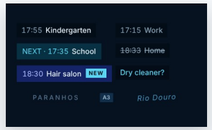


MAP

Car marker

Moving, with heading

S02 to S19, C4, V2



MAP

Map label

Upcoming · next · new · done · cancelled · suggested. Place names and road shields

S02 to S20, C4, V2

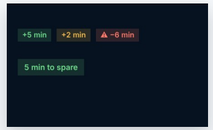


CONTENT

Panel

Narrow · wide · form. Glass at 30 to 60 percent, 24 px blur, no border

S02 to S20, C1, C4, C5, V2

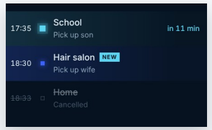


CONTENT

Spare-time pill

Green · amber · red with minus and warning mark · large

S09, S12 to S17, S20, V2

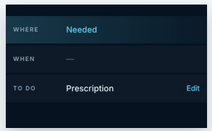


CONTENT

Stop row

Upcoming · next · arriving · done · summary · new · changed · cancelled · dragging

S09, S12 to S15, S17, S19, C1, V2

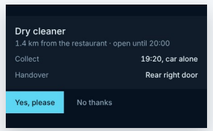


CONTENT

Briefing slot

Empty · asked · filled · editable

S04 to S07

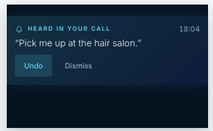


CONTENT

Suggestion card

Florist · errand, with key–value rows and a quoted message

S10, S18

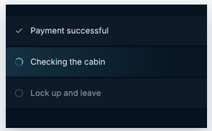


CONTENT

Change banner

Automatic · heard phrase with chime and Undo · running late with fixes

S09, S17, V2

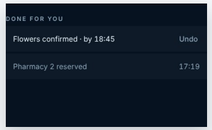


CONTENT

Step list

Done · running · waiting

S11, S19, C1, C2

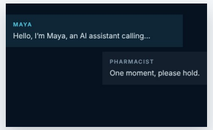


CONTENT

Ledger row

Latest with Undo · older with time

S14, S16

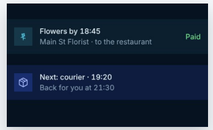


CONTENT

Call transcript

Maya · other party, hold timer, outcome card

S08

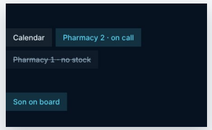


CONTENT

Info row

Icon tile, title, detail and status

S12, S20

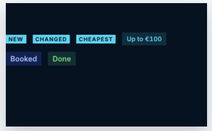


CONTENT

Chip

Default · active · failed · state

S02, S03, S08, S14, S16



CONTENT

Badge

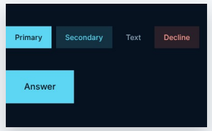
New · changed · cheapest · booked · done · limit

S03, S09, S17, S19, C4, V2

COMPONENT INVENTORY · 2 OF 2

Inputs, overlays, the phone and the strip.

Specimens drawn with the same code as the screens.
Every tap area is at least 76 × 76 px.

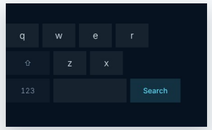


INPUTS

Button

Primary · secondary · text · destructive · call

All screens



INPUTS

Keyboard

Letter · modifier · space · action key

S05

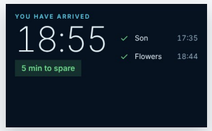


OVERLAYS

Intent card

Action, detail, progress and what comes next

S13

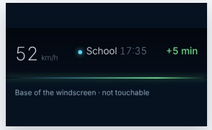


OVERLAYS

Arrival report

Large time, pill, what was done, next step

S20

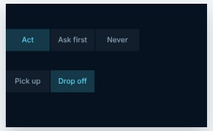


PHONE AND STRIP

Strip band

Presence · next stop · spare time · progress · event · horizon line in cyan, green, amber, red · offline · off

All strip states

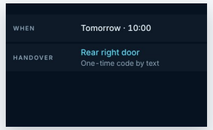


INPUTS

Segmented control

Two or three options

S03, C5

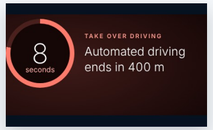


INPUTS

Form row

Label and value · with hint · selected

C5



OVERLAYS

Takeover alert

Countdown ring, red, Level 3/4 variant only

V1



OVERLAYS

Car top view

Default · selected compartment · door open · cabin scan

C1, C2, C5

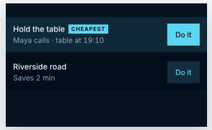


INPUTS

Toggle

On · off

S03

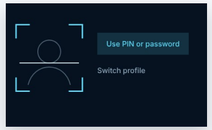


INPUTS

Option row

Recommended · default, each with an action

V2

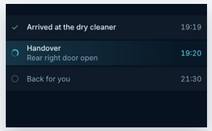


OVERLAYS

Sign-in

Face scan · PIN or password · switch profile

S01

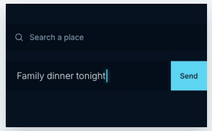


PHONE AND STRIP

Phone run

App bar, live map card, steps: done · running · waiting

C2, C3

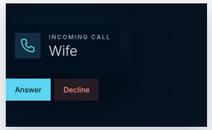


INPUTS

Text field

Search · free text with caret · with Send

S05, S06

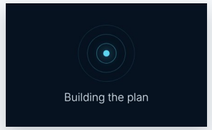


OVERLAYS

Incoming call

Answer · decline

S16



OVERLAYS

Processing

“Building the plan” with sonar rings

S07



PHONE AND STRIP

Camera still

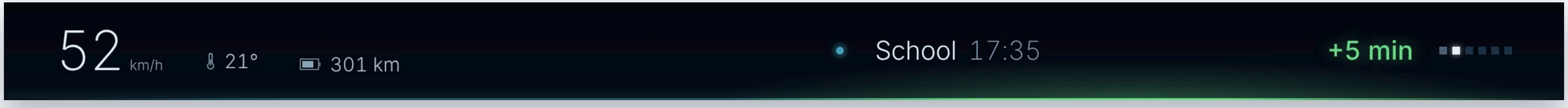
Detection frame, camera label, result chip

C3

LED STRIP · BASE OF THE WINDSCREEN · NOT TOUCHABLE

One band, always dark, nothing unique.

Never on the strip: transcripts, maps, editing, anything long or interactive, medication names, personal data.



1 Automotive half
Speed, cabin temperature and range. Always on the left, always the same.

2 Presence
A small dot: listening, hearing you, speaking (rings), thinking (hollow), on a call (phone).

3 Next stop
Name and planned time. Replaced by an event for 4 seconds.

4 Spare time
The figure and the line share one colour: green, amber or red.

5 Progress
One square per stop: grey done, white current, dark to come.

6 Horizon line
Cyan under the car data, the spare-time colour under Maya. Flashes white on events.

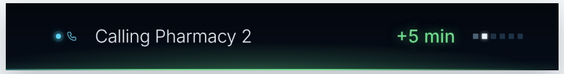
STATES THROUGH THE STORY · MAYA'S HALF

S02 Signed in



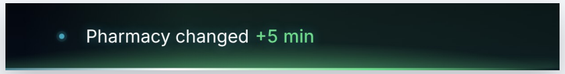
No plan yet: neutral line, no spare time.

S08 Pharmacy call



On a call. The plan exists, so spare time appears.

S09 Event · plan changed



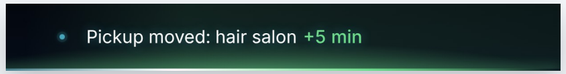
A change in the world: no chime, the line flashes white for 4 s.

S12 Plan running



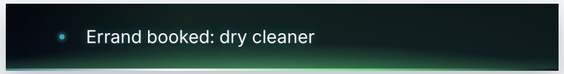
Next stop, spare time, one square per stop.

S17 Event · ambient change



Heard in the call: the chime sounds once, the strip flashes.

S19 Event · errand booked



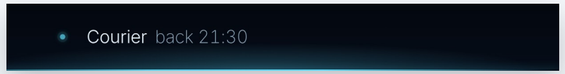
Agreed with the shop. Spare time unchanged.

S20 Arrival



Every square filled; the final spare time.

C1 Courier hand-off



What happens next and when the car is back.

C2 Off · cabin empty



Nobody inside: the whole band is off.

Edge 2 Amber · 0 to 4 min



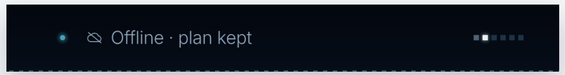
The number is always shown; the line turns amber.

V2 Red · below 0



A minus sign and a warning mark, never colour alone.

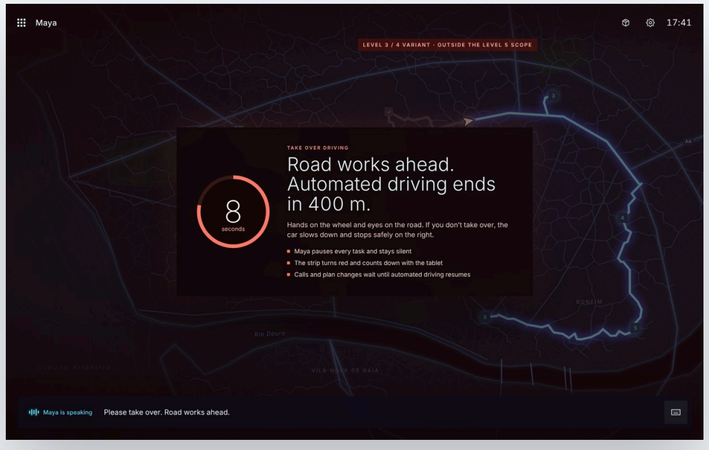
Edge 8 Offline · plan kept



A grey dotted line; the plan keeps running.

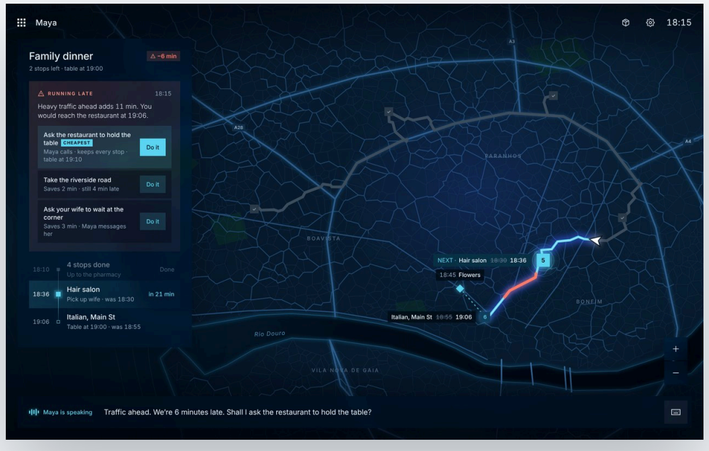
VARIANTS AND EDGE CASES

Seventeen ways the evening could go wrong, and two of them drawn.



V1 Takeover request

Level 3 or 4 only, outside the Level 5 scope: an 8-second countdown on tablet and strip; Maya pauses every task and stays silent.



V2 Running late · edge case 2

−6 min with a warning mark. Three fixes, cheapest first, each with Do it. Maya speaks once, because the promise is at risk.

#	EDGE CASE	EXPECTED BEHAVIOUR
1	No pharmacy on the route has stock	Widen the search, show it on screen; ask only if the detour breaks spare time
2	Spare time goes negative	Fixes, cheapest first, such as holding the table (V2); the strip turns red
3	Ambient false positive: a child says “let’s go to the park”	Act only when confident, otherwise ask; chime, heard phrase and Undo
4	The chime is missed	The change stays in the ledger, and the strip repeats it once
5	The florist can’t deliver before 19:00	Try the next florist; otherwise offer a pickup stop
6	A pharmacist refuses an AI caller	Next pharmacy, with the refusal logged
7	A child is not at the pickup point	Alert on the tablet, a wait window, a call to the school
8	Connectivity loss	Keep the current plan; the strip shows Offline; edits queue
9	Dinner time or place changes by message	Same pattern as the wife’s call: chime and Undo
10	Several people speak	Commands only from the signed-in user; other speech is context
11	Shop not ready, or nobody comes to the car	Wait within a set window, then leave to be back on time; the report says why, the errand returns to Errands
12	Wrong item loaded, or more than one	The camera checks before the door locks; Maya texts the shop and waits within the window
13	Someone or something left in the cabin	A person or a pet stops courier mode; a bag gets “Please take your belongings” and a note on the phone
14	The car is needed earlier	Michael calls it; the same car comes from charging, or finishes the handover first
15	Return time at risk	Tell Michael early on the phone, with the new time; never late in silence
16	Wrong door, or the code is shared	The code opens only the agreed door, once, at the agreed place and time
17	Item not allowed: too big, an animal, needs a signature	Declined while booking, before anything is promised

Cases 11 to 17 cover courier mode. Cases 1, 3 and 4 also appear in the prototype.

VALIDATION AND METRICS

Riskiest first: does ambient intent feel helpful, or creepy? Would people send an empty car?

TESTS

TEST	METHOD	SUCCESS SIGNAL	CHECKS
Ambient-intent false positives	Scripted audio clips, true changes and decoys; measure Undo and reaction	Few false positives; chime and phrase understood	A3, A9
Courier trust test	Clickable courier flow, S18 to C3, 5 participants: what would they send, what would stop them?	4 of 5 would send it for an everyday errand	A14
Glance test	5-second exposure: “What’s next? How much spare time?”	80% or more correct	A8, A10
Replan legibility	Show a change, ask “what changed?”	Understood in under 5 s	A9
Touch-only run	Brief and edit a plan without speaking	Same completion as voice	A11
Wizard of Oz briefing	A person plays Maya’s voice, 5 participants (if I had a month)	No confusion; count the questions	A1, A6

The glance, legibility, touch-only and courier trust tests can run remotely on the clickable prototype. Scripted calls to real shops (A4, A13) and the Wizard of Oz session are on the one-month list.

SUCCESS METRICS · ILLUSTRATIVE TARGETS

On-time arrival, within the 5 minutes of spare time	**95% or more**
Replans that keep spare time at or above zero	**90% or more**
Questions Maya could have answered herself	**Under 10%**
Undo rate on automatic and ambient changes	**Under 10%**
Time from a change to a visible adapted plan	**3 s or less**
Proactive suggestions accepted	**50% or more**
“Feels like a great assistant”	**4 / 5 or more**
Courier errands completed without Michael stepping in	**90% or more**
Car back by the promised time after a courier run	**95% or more**

ASSUMPTIONS · HIGH RISK IN RED

A1	Users accept “do first, show clearly, undo easily”
A2	Maya can predict the destination and learn the day from habit and data
A3	Listening to cabin talk and calls is welcome; false positives are cheap
A4	Retailers cooperate with a self-identified AI caller
A5	Automatic payments from Maya Wallet, up to a limit, are acceptable
A6	Users accept as many questions as needed, if each is necessary
A7	A suggestion tied to a known occasion is welcome
A8	Spare time gives at-a-glance confidence
A9	Chime, heard phrase and Undo are understood in a noisy cabin
A10	A passive strip carries enough at a glance
A11	Voice and touch parity is achievable for every task
A12	On-time arrival is the main success criterion
A13	Shops will hand items to an empty car that opens one door with a code
A14	People will send their empty car, and accept a cabin camera
A15	An empty Level 5 car may legally run errands

TRADE-OFFS AND WHAT'S NEXT

What each choice gains, what it costs, and what I would do with a month.

DECISION	GAIN	COST
Level 5 only	One calm, trip-level experience	No driver-state design; V1 sketches Level 3 or 4
Full trust: Maya acts, pays and listens	Feels like an executive assistant	No approval before calls, or payments under the limit; trust lives in S03 and Undo
Maya speaks rarely	A quiet cabin	Voice-only users get less status; captions and the strip help
Ambient intent: act when sure, ask when unsure	Plans follow life without being managed	False positives, answered by chime, heard phrase and Undo
Silent retailer calls, shown as transcripts	Silence in the cabin	Michael can read what Maya promised, not hear it
Banners stay until dismissed	Nothing is lost	Clutter if changes pile up
Undo on the latest ledger row only	A calm ledger, one clear action	Older actions are slower to reverse
Edit plan with a ripple preview, then Apply	Informed edits	An extra step for a quick change
A passive strip with nothing unique	No second interactive surface	The strip can never be the only channel
One layout: plan left, flat map	Learnable and consistent	No right-to-left variant, no 3D context
Courier: no handoff, no commitment	Errands don't fail at the door	Fewer errands qualify
Courier: one door, one code, a cabin camera	The rest of the car stays closed	A camera in the cabin; the item must fit the compartment
A solo exercise, no user research	Speed	A1 to A15 are untested

STILL OPEN

When Maya is unsure about a payment: ask first, or act and show Undo? Leaning towards act and Undo, below a limit.

SCREENS FOR LATER

The app launcher's contents (the icon stays), creating a new profile, idle or standby when nobody is signed in, guest mode.

IF I HAD A MONTH

- Wizard of Oz sessions and a study of real family trips
- Scripted calls to real shops, for AI calls and empty-car handovers
- Other errand types beyond pick-ups and drop-offs
- Responsive layouts: each screen size designed, not scaled
- Trust that grows over time, a full activity log, wallet top-ups and statements
- Multi-passenger conflicts and escalation to a human
- Accessibility research with touch-only and voice-only users
- A regulation review for empty cars (A15)